Child Marriage, Legal Regimes, and Life Outcomes

Theoretical Framework and Empirical Evidence

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Abstract

This paper studies how minimum-marriage-age laws shape women's outcomes in sub-Saharan Africa. We develop a three-stage model linking legal enforcement, pre-market educational investment, marriage-market selection, and intra-household bargaining. In the model, stricter enforcement delays marriage, later marriage increases the human capital women bring into marriage, and these entry conditions affect household outcomes through selection and bargaining. We test the model using a regression discontinuity design across twelve African border pairs, comparing women who live near the same international border but face different legal regimes. Women on the stricter side of the border complete 0.33 additional education levels, have their first birth 0.93 years later, have 0.33 fewer children, and are 2.9 percentage points more likely to work in white-collar employment; the continuous years-of-education measure also rises by 1.21 years, though this estimate is markedly less stable than the other results once we exclude individual border pairs or the Namibia sub-sample. Taken together, the results suggest that minimum-marriage-age laws affect women's lives not only by delaying marriage directly, but also by changing pre-marital investment and the conditions under which women enter household bargaining.

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I. Introduction

Marriage has been central to social organisation for as long as there have been societies to organise, yet economists largely did not treat it as an economic institution until Becker (1973, 1974) offered a framework for treating it as something other than a cultural given. Since then, the economics of marriage has grown substantially, absorbing insights from matching theory, household bargaining, and human capital accumulation (Wahhaj, 2018). Still, a narrower question about what happens to women's lives when they marry young, and what role law plays in enabling that, has not received enough empirical attention, particularly in sub-Saharan Africa where the practice remains widespread. This paper tries to fill part of that gap, by studying child marriage as both a market equilibrium outcome and a policy object.

The paper has a theoretical and an empirical part. A large body of empirical work documents that women who marry young fare worse on a range of different outcomes: education, fertility, employment, health (Field and Ambrus, 2008; Wilson, 2022). What that work generally cannot tell us is why. Does early marriage truncate schooling directly? Does it shift bargaining power inside the household? Does it change the composition of who enters the marriage market at later ages, and in ways that feed back into individual decisions? Without a formal framework, these mechanisms are difficult to distinguish and difficult to test separately. The model serves an organisational purpose: it makes precise which margins the legal environment should move, in what order, and why, so that the empirical results can be evaluated against specific predictions rather than post-hoc rationalisations.

The model connects three objects: the legal regime, the marriage market, and the married household. A girl's family chooses how much to invest in her education before she enters the marriage market, knowing that this investment will be worth more the older she is when she marries, because the returns to education are convex in time. Once she enters the market, the legal minimum age operates as a cost on underage marriage. A stricter regime lowers marriage hazards before the threshold, which delays marriage and changes who remains unmarried at later ages. That process of selection feeds directly into the household problem, because the state a woman carries into marriage (her education, her human capital, the assets she brings) determines how that household will operate. The household problem is specified explicitly, and its value function is what enters the pre-marriage value of accepting a match. The model is therefore recursive: legal enforcement shapes the distribution of marriage timing, timing shapes household entry conditions, and those entry

conditions shape the downstream outcomes we observe in the data.

The empirical strategy is designed with this chain in mind. We use Demographic and Health Survey data from twelve border pairs in sub-Saharan Africa, where two neighbouring countries share a land border but differ in their minimum legal marriage age. The border creates a sharp discontinuity in the legal regime while holding a great deal else constant. We estimate a regression discontinuity design following the border design of Anderson (2018), using residence on the high-age side of the border to identify the intent-to-treat effect of the legal environment on women's outcomes. The estimand is the total causal effect of the child marriage legal environment on women's outcomes. The enforcement cost of underage marriage is the primitive that differs across the border, and it affects outcomes through two simultaneous channels: realized marriage timing and anticipatory pre-market investment by families who expect their daughters to marry young. Our main results are the reduced-form effects of residing on the strict side. We additionally test whether legal exposure moves the child marriage rate itself, which lets us confirm that the legal discontinuity is a real behavioural margin rather than treating that relationship as an instrument to rescale the other outcomes.

The reduced-form results are consistent with the model's predictions. Women on the strict side of the border have completed higher levels of education and are more likely to have received any schooling at all, relative to women on the lenient side. Age at first birth increases, children ever born decreases, and age at first sex is delayed. White-collar employment rises on the extensive and intensive margins. Broad labour-force participation also shifts: women on the strict side are more likely to report not working, though the model treats this margin as theoretically ambiguous.

The contribution is a recursive framework in which legal enforcement, pre-market investment, marriage market selection, and household bargaining are connected by a single structure, paired with an empirical strategy that uses legal discontinuities at international borders to provide plausibly causal evidence under the border-smoothness assumption.

The remainder of the paper proceeds as follows. Section II reviews the related literature on child marriage, female legal rights, and intra-household bargaining. Section III develops the three-stage theoretical model. Section IV describes the data sources and sample construction. Section V specifies the empirical strategy and identification assumptions. Section VI reports the main estimates and heterogeneity analysis. Section VII presents the robustness checks. Section VIII discusses the interpretation of the findings, limitations, and directions for future research. Section IX concludes.

II. Literature Review

We organise the review around the theoretical and empirical mechanisms used in this study: marriage timing as an economic decision, human-capital accumulation before and around marriage, legal enforcement, household bargaining, and empirical identification.

A. Marriage Timing as an Economic Decision

The systematic study of marriage as an economic institution begins with Becker (1973) and Becker (1974), who modelled the marriage market as an assignment problem in which individuals sort on the basis of comparative advantage and complementarities in household production. The central insight is that marriage generates surplus through specialisation and risk-sharing, and that who marries whom, and when, is the outcome of a competitive equilibrium rather than a social convention. Subsequent work formalised this using tools from matching theory. Shimer and Smith (2000) extended the framework to allow for search frictions, establishing conditions under which positive assortative mating arises when match quality is complementary. Our model contains this search structure: women meet potential grooms with exogenous probability and marriage occurs when the match surplus is non-negative after accounting for the legal wedge.

A related strand asks why marriage timing varies systematically with education and economic conditions. Iyigun and Lafortune (2016) propose a model in which educational and marriage decisions are jointly determined, showing that minimum age laws can raise attainment by delaying the point at which marriage competes with schooling. Our model formalises such a complementary mechanism as returns to pre-market investment are convex in the age at marriage; early marriage depresses both investment and attainment through two reinforcing channels.

The marriage market also responds to economic incentives and local institutions. Corno et al. (2020) study how income shocks affect the timing of child marriage in sub-Saharan Africa and India, finding that droughts raise early marriage hazards in bride price societies through a consumption-smoothing motive. Their paper establishes that marriage timing in the region responds to economic incentives in ways consistent with an equilibrium model, which supports treating marriage timing as an optimising outcome rather than just a cultural one.

Wahhaj (2018) is the direct theoretical predecessor to our model in the child marriage literature. His framework captures how parental preferences and social norms sustain early

marriage in equilibrium, but does not embed a solved household problem into the continuation value of marriage.

B. Child Marriage and Human Capital Accumulation

The empirical literature most directly related to our work studies the consequences of early marriage for women's human capital and later outcomes. Field and Ambrus (2008) is the closest antecedent. They exploit variation in age at menarche as an instrument for age at first marriage in rural Bangladesh, finding that delaying marriage by one year increases female educational attainment by 0.22 years.

The connection between marriage matching and human capital investment is developed by Chiappori et al. (2017), who embed marriage matching into a transferable utility framework and show that rising returns to education reinforce assortative matching on schooling.

Jayachandran and Lleras-Muney (2009) provide supporting evidence by showing that longer expected life horizons raise educational investment in developing countries, because the returns to schooling accumulate over time.

The marriage market also interacts with cultural institutions that shape investment in daughters. Ashraf et al. (2020) show that the bride price custom creates incentives for parents to educate their daughters, since a more educated daughter commands a higher transfer at marriage. School construction programs in Indonesia and Zambia raised female schooling significantly among ethnic groups with the bride price custom but had no effect among groups without it.

Heath and Mobarak (2015) study how the growth of Bangladesh's garment sector changed girls' schooling, work, marriage, and fertility decisions. Girls exposed to garment-sector opportunities delay marriage and childbirth and are more likely to remain in school at younger ages, while older girls are more likely to engage in wage work. Their evidence supports the broader mechanism that marriage timing responds to the economic returns to female education and work: when opportunities outside early marriage improve, schooling becomes more valuable and early marriage becomes more costly. Duflo et al. (2015) demonstrate a related schooling-cost channel in a seven-year randomized evaluation in Kenya, showing that education subsidies reduce dropout, pregnancy, and marriage, while an HIV curriculum stressing abstinence until marriage does not reduce pregnancy or STI.

C. Legal Regimes and Enforcement

The strand of work closely related to the policy object in this paper studies how legal regimes governing marriage and family law change women's outcomes. The central insight across this literature is that legal regimes affect behaviour not only through direct enforcement but through the outside options and expectations they create. A woman's position inside a marriage depends on what exit from that marriage looks like, and the legal rules that shape outside options matter even for women who never exercise them.

Wilson (2022) uses legislative reforms raising the minimum legal marriage age across seventeen low and middle-income countries as natural experiments, finding significant positive effects on female schooling and labour market participation. Our approach differs from Wilson's in two important respects. First, our identification strategy uses cross-border variation in legal regimes rather than within-country pre/post variation, addressing the concern that legislative reforms are themselves endogenous to trends in women's outcomes. Second, our estimates are grounded in a theoretical model that specifies the mechanisms through which legal enforcement operates, so the results carry interpretive content beyond just the reduced-form results.

Collin and Talbot (2023) address a closely related but distinct question: whether statutory minimum-age laws are binding in practice. Using regression-discontinuity methods on DHS and MICS data from over one hundred countries, they test whether marriage rates change discontinuously at each country's legal cutoff. They find limited evidence that minimum-age laws are consistently enforced, and argue that legal reform is more likely to matter when paired with monitoring and enforcement capacity. Their findings motivate our interpretation of the legal regime as an enforcement-relevant environment rather than merely a statutory rule on paper. They also provide context for our urban-rural heterogeneity analysis: if formal institutions are more accessible in urban areas, the legal discontinuity may be more binding there.

Buchmann et al. (2023) use a field experiment in Bangladesh to test whether financial incentives and girls' empowerment programs can delay marriage. They find that girls eligible for a financial incentive for at least two years are 19 percent less likely to marry underage, while the empowerment program did not reduce adolescent marriage. This evidence is consistent with the broader premise of our model that marriage timing responds to incentives and constraints in the marriage market, although their intervention operates through financial incentives rather than legal enforcement.

Legal rules also provide one source of exogenous variation in marriage timing. Dahl (2010) uses variation in US state laws governing minimum marriage age, compulsory schooling, and child labour as instruments for early teen marriage and schooling choices, finding that women who marry young are substantially more likely to live in poverty as adults.

Voena (2015) estimates a structural model of household decision-making, showing that divorce laws affect couples' intertemporal choices, with effects that depend on the property-rights regime governing marital assets. This forward-looking response to family law is related to the logic of Stage 1 in our model, though the setting differs: in our framework, families' education investment responds to the expected marriage-timing environment rather than to divorce risk. Fernández and Wong (2014) provide complementary evidence that changes in the risk of marital dissolution can affect women's labour-supply decisions, finding that higher divorce probability and changes in the wage structure can each explain a large share of the rise in married women's labour force participation in the United States.

D. Household Bargaining, Fertility, and Labour Supply

Our married-household problem draws on the collective household model of Chiappori (1988) and Chiappori (1992). The core idea is that household members may have distinct preferences, but household choices can be represented as Pareto-efficient allocations governed by a sharing rule or Pareto weights. Chiappori et al. (2002) extend this framework by showing how distribution factors, including sex ratios and divorce laws, can shift household members' bargaining positions without directly changing preferences or the joint budget set. Our model applies this logic to marital entry states: in our framework, women who enter marriage with more education and human capital have stronger outside options, which raises their Pareto weight and shifts household choices over fertility and labour supply.

Ashraf et al. (2014) test the role of household bargaining in fertility decisions using a field experiment in Zambia. They vary whether married women receive access to contraceptives alone or in the presence of their husbands, and find that women assigned to receive access with their husbands are less likely to seek family-planning services, less likely to use concealable contraception, and more likely to give birth. Their results show that fertility choices can reflect bargaining frictions and asymmetric information within marriage, rather than only a single household preference. This mechanism is related to Lemma 4 of our model, where lower female bargaining weight expands the region in which birth is chosen.

Our contribution is to connect the source of bargaining power at marital entry, including education and human capital shaped by child-marriage legal regimes, to the fertility and labour-supply outcomes studied in the household-bargaining literature.

Garcia-Hombrados (2022) uses age discontinuities in exposure to Ethiopia's Revised Family Code to estimate the causal effect of delaying teenage cohabitation on infant mortality. He finds that a one-year delay in age at cohabitation reduces the probability that a woman's first-born child dies during infancy by 3.8 percentage points, with the mechanism closely linked to the resulting delay in age at first birth.

E. Identification: Legal Variation and Border Designs

The empirical challenge in this literature is selection. Women who marry young differ from those who marry later in ways that are correlated with almost every outcome of interest. Naive comparisons confound the effect of marriage timing with the characteristics of women who marry at different ages. Credible identification therefore requires variation in marriage timing that is generated by legal, biological, institutional, or policy variation rather than by the woman's own characteristics or family preferences.

Reform-based designs provide one such source of variation. The approach in Wilson (2022), discussed above, uses spatial and cohort variation in exposure to child-marriage bans in a difference-in-differences framework. Relative to reform-based designs, our approach uses cross-border discontinuities in legal regimes rather than within-country exposure to legislative reform. This shifts the identifying comparison from changes over time within reforming jurisdictions to nearby populations on opposite sides of a border that face different legal marriage-age environments.

The closest methodological antecedent to our empirical strategy is Anderson (2018), who uses variation in colonial legal origins across African borders as an instrument for gender-related legal institutions, studying their effects on female HIV prevalence. Anderson's border design is the foundation of our own identification approach. We follow their empirical design by using GPS-linked individual microdata from the DHS, which allows each survey cluster to be located relative to the legal cutoff and enables a regression discontinuity framework.

The framework that underpins our design rests on methodological foundations laid out by Imbens and Lemieux (2008) and Lee and Lemieux (2010). These papers establish the conditions under which regression discontinuity designs identify treatment effects at the

cutoff, and set out the diagnostic tests that allow researchers to assess whether the identifying assumptions hold in practice.

F. Gaps this Paper Addresses

Taken together, the existing literature leaves two gaps this paper addresses. The first is theoretical. Existing work explains marriage as an equilibrium outcome, shows that marriage timing responds to incentives, and studies how bargaining power affects household choices. What is less developed is a single framework connecting the legal minimum marriage age, pre-market educational investment, marriage-market selection, and intra-household bargaining. Relative to Wahhaj (2018), our model keeps the focus on early marriage as a marriage-market outcome but extends the analysis to the household continuation problem and to downstream outcomes through marital entry states. It therefore treats the legal regime, the marriage market, and the married household as connected parts of a single dynamic problem rather than as separate stages.

The second gap is empirical. The literature has made significant progress in credibly estimating particular effects of child marriage in specific countries and settings, including schooling in Bangladesh (Field and Ambrus, 2008), schooling and labour-market outcomes across multiple countries (Wilson, 2022), and infant mortality in Ethiopia (Garcia-Hombrados, 2022). What is less common is a single empirical design that tests the joint predictions of a recursive model. We estimate effects on schooling, fertility and skilled employment from the same border discontinuity and the same sample, which lets us read the pattern of outcomes against the model's predictions rather than evaluating each margin in isolation.

III. Theoretical Framework

The model proceeds in three stages. Stage 2 (the marriage market) is a reduced-form marriage-market block inspired by Wahhaj (2018). It preserves the key insight that delaying marriage is costly because future match opportunities are uncertain, but abstracts from Wahhaj's asymmetric-information and reputation structure to focus on the legal wedge and its downstream consequences. Stages 1 and 3 are the extensions introduced in this paper. Stage 1 is a pre-market education investment problem in which a girl's family chooses schooling investment before she enters the marriage market: when early marriage is anticipated, the expected returns to education are lower, depressing investment before any

marriage occurs. We formalise this through a convex accumulation function and embed it in the recursive structure of the model. Stage 3 is a married-household bargaining problem in which the woman's entry state (her education and human capital at the time of marriage) determines her bargaining weight and the household's subsequent choices over fertility, labour supply, and schooling. The stages are numbered in the order experienced by the woman. The family's Stage 1 investment decision responds to the expected Stage 2 marriage-market equilibrium; the Stage 2 outcome shapes the entry state carried into Stage 3; and the Stage 3 value of that entry state is what Stage 1 families anticipate when choosing κ.

A. Setup

Two jurisdictions are indexed by $j \in \{\text{Strict}, \text{Lenient}\}$, corresponding to the two sides of an international border. Residence is fixed and determines the applicable legal regime. The minimum legal age for female marriage in jurisdiction j is $\bar{a}_j$, with $\bar{a}_{\text{Strict}} = 18$ and $\bar{a}_{\text{Lenient}} < 18$. The only primitive that differs across jurisdictions is the expected utility-equivalent cost of contracting an underage marriage. For a woman of age a, the enforcement wedge $\Lambda_j(a)$ is positive for $a < \bar{a}_j$ and zero otherwise:

$$\Lambda_j(a) = \mathbf{1}\{a < \bar{a}_j\}\ell_j(a), \quad \ell_j(a) \geq 0, \tag{1}$$

where $\ell_j(a)$ captures the expected utility-equivalent cost of enforcement (invalidation risk, administrative barriers, anticipated sanctions), which may vary by age and jurisdiction.

Assumption 1 (Single-primitive legal variation)**.** *All non-policy primitives (preferences, technology, non-policy institutions) are identical across jurisdictions.*

Bargaining primitive We introduce here a single bargaining primitive $\theta \in (0,1)$ that will appear in both Stage 2 and Stage 3. In Stage 2, θ governs the fraction of her gross marriage gain that the woman retains after intra-household frictions. In Stage 3, θ enters the Pareto weight $\omega = \Omega(b_0, h_0, e_0; \theta)$ that determines how household preferences aggregate. The same θ appears in both roles by assumption: it is a single deep parameter of the woman's bargaining strength, and we do not derive the link between its Stage 2 and Stage 3 manifestations from a more primitive bargaining protocol. The mapping $\Omega(\cdot;\theta)$ is treated as maintained throughout, with weak monotonicity in each argument; the micro-foundation is discussed in Appendix A.

B. Stage 1: Pre-Market Education Investment

Before entering the marriage market at age a_0, a girl's family chooses an education investment $\kappa \in [0, \bar{\kappa}]$ at convex cost $c(\kappa)$, with $c' > 0$ and $c'' > 0$. Let a_s denote the school-starting age and $T_m = a_m - a_s$ the years of potential schooling before marriage. The educational endowment at marital entry is

$$e_0(T_m; \kappa) = e_b + \kappa \cdot T_m^{\gamma}, \quad \gamma > 1, \tag{2}$$

where $e_b \geq 0$ is a baseline education level and κT_m^{γ} is the family-investment component. The parameter $\gamma > 1$ imposes increasing returns to schooling exposure: each additional year before marriage builds on skills accumulated in prior years, making the return strictly convex in T_m. This is consistent with dynamic complementarity in skill formation (Ben-Porath, 1967; Cunha and Heckman, 2007): human capital is self-productive, so earlier investments raise the productivity of later ones. Early marriage truncates accumulation before girls reach the high-return portion of the schooling path.

The family forms expectations over the equilibrium distribution of a_m under the applicable legal regime j, noting that $M(a_m; \kappa)$ (defined in Stage 2 below) already integrates over idiosyncratic draws ξ internally. The jurisdiction-specific investment problem is

$$\kappa_j^* = \arg \max_{\kappa \in [0, \bar{\kappa}]} \mathbb{E}_{a_m | j} \big[M(a_m; \kappa) \big] - c(\kappa), \tag{3}$$

where the outer expectation is taken only over the equilibrium distribution of marriage ages under regime j. Here M is the daughter's expected value from the marriage–household continuation problem; the family maximises her individual expected welfare rather than a joint household objective. The first-order condition is

$$\mathbb{E}_{a_m | j} \left[\frac{\partial M}{\partial e_0} \cdot T_m^{\gamma} \right] = c'(\kappa_j^*), \tag{4}$$

The sign $\partial M / \partial e_0 > 0$ is motivated by two channels: the direct effect of e_0 on the wage w^f and on the continuation V_2 (the latter via Assumption 6 combined with $\partial e_1 / \partial e_0 = 1$), and the weak monotonicity of the Pareto weight $\omega = \Omega(b_0, h_0, e_0; \theta)$ in e_0; at any optimal household choice d^*, both channels move the woman's value M in the same direction. Because $M(a; \kappa)$ is itself an upper envelope of $\{\Pi_f, \Pi_s, \Pi_\ell\}$, and V_2 and Ω are maintained rather than derived primitives (Assumption 6), these two channels are sufficient conditions for, but do not by themselves establish, the curvature and cross-partial properties needed

for the Stage 1 comparative statics below. We state those properties directly as a maintained regularity condition on M, in the same spirit as the maintained treatment of V_2 and Ω.

Assumption 2 (Regularity and complementarity of the marriage-market value function)**.** *For every $a \in \mathscr{A}$ and jurisdiction j: (i) $M(a;\kappa)$ is continuously differentiable and strictly concave in κ on $[0,\bar{\kappa}]$; and (ii) M has strictly increasing differences in (κ,a): for $a' \geq a$ and $\kappa' \geq \kappa$,*

$$M(a';\kappa') - M(a';\kappa) \;\geq\; M(a;\kappa') - M(a;\kappa),$$

with strict inequality whenever $a' > a$ and $\kappa' > \kappa$. Condition (ii) is motivated directly by $\partial^2 e_0/\partial\kappa\,\partial T_m > 0$ together with the two channels above, and formalizes the economic content of Proposition 1: the value of additional investment is higher for women who marry later.

Proposition 1 (Education investment and marriage-age exposure are complements)**.** *For any fixed κ, the expected marginal return to education investment is strictly increasing in expected marriage-age exposure $\mathbb{E}[T_m \mid j]$. If the strict regime induces a weak upward shift in the distribution of T_m, then $\kappa^*_{Strict} \geq \kappa^*_{Lenient}$, with strict inequality if the shift is strict over a set of positive probability and the optimum is interior. The complementarity condition is*

$$\frac{\partial^2 e_0}{\partial\kappa\,\partial T_m} = \gamma T_m^{\gamma-1} > 0,$$

so the marginal return to each unit of investment κ is strictly increasing in schooling exposure T_m. Families facing shorter expected schooling horizons under lenient regimes therefore rationally invest less. Stage 2 provides the mechanism by which stricter enforcement shifts the distribution of marriage ages upward. (Proof in Appendix A, using Assumption 2)

Remark 1. With convex returns ($\gamma > 1$), the marginal return to additional schooling exposure is highest at longer durations. Early marriage removes those years, truncating accumulation where the curve is steepest. The lenient legal regime therefore depresses educational attainment through two channels: realised truncation from lower a_m in equilibrium, and the anticipatory reduction in κ that follows from lower expected marriage age.

C. Stage 2: The Marriage Market

Time is discrete and indexed by age $a \in \mathscr{A} \equiv \{a_0, a_0+1, \ldots, a_1\}$, where a_0 is entry into the marriage market and a_1 is the terminal age. At age a, an unmarried woman meets a potential groom with probability $\lambda(a) \in (0,1)$, weakly decreasing in age: meeting rates are

highest at younger ages when the pool of unmatched men is largest, and fall as the cohort thins. A woman who declines a match at age a faces a lower probability of receiving an offer at $a+1$, which is why early marriage can be an equilibrium outcome even when later marriage yields a higher entry state.

Let $M(a;\kappa_j^*)$ denote the woman's expected lifetime value from the household continuation problem, conditional on marrying at age a:

$$M(a;\kappa_j^*) = \mathbb{E}_\xi\left[U_1^f(S_0, d^*(S_0,\omega);\omega) + \beta V_2(S_1)\right], \tag{5}$$

where U_1^f is the woman's period-1 flow utility at the household's optimal choice $d^* = (f_1^*, s_1^*, \ell_1^*)$, and $V_2(S_1)$ is the period-2 continuation value evaluated at the updated state $S_1 = (h_1, e_1, n_1)$. M is the woman's value at the equilibrium household choice; it differs from her post-bargaining marriage-market payoff $U_j^f(a)$, which adds the surplus-split adjustment described below. The woman's gross gain from accepting a match at age a, net of the enforcement cost and her outside option, is

$$\Delta_j(a;\kappa_j^*) = M(a;\kappa_j^*) - \Lambda_j(a) - V_j(a+1;\kappa_j^*). \tag{6}$$

The woman accepts the match if and only if $\Delta_j(a) \geq 0$.

Assumption 3 (No indifference, generically in parameters)**.** *Fix the functional forms of V_2, Ω, w^f, Λ_j, $\lambda(\cdot)$, and $c(\cdot)$, and let Θ denote the finite-dimensional space of the model's remaining parameters (including θ, γ, e_b, υ_m, υ_f, χ_s, χ_ℓ, ϕ_s, ϕ_ℓ, R, and y_m). For Lebesgue-almost every parameter vector in Θ, every equilibrium fixed point μ^{j*} (in the sense of the existence proof in Appendix A) satisfies $\Delta_j(a;\kappa_j^*(\mu^{j*})) \neq 0$ for every $a \in \mathscr{A}$ and every jurisdiction j. This is a genericity condition on the model's primitives rather than a condition imposed directly on the endogenous equilibrium object: it rules out a knife-edge coincidence between one of the finitely many ages in $\mathscr{A}$ and the zero set of $\Delta_j(\cdot;\kappa_j^*(\mu^{j*}))$, which is a measure-zero event in parameter space for any fixed $\mathscr{A}$. It avoids randomised tie-breaking and ensures that the equilibrium marriage-age distribution is uniquely determined by the Stage 2 backward induction.*

Assumption 4 (Male participation)**.** *The man's value from the match weakly exceeds his outside option whenever $\Delta_j(a) \geq 0$, and weakly exceeds his outside option throughout the marriage continuation thereafter. The feasibility constraint therefore binds only for the woman, both at acceptance and during the household problem, and $\lambda(a)$ is purely the probability of a meeting. In settings where men also face legal exposure (through sanctions*

on grooms or their families), the feasibility constraint could bind for both parties; we abstract from this channel.

Under Assumption 4, the woman is the only party whose participation can bind, and the exit conditions defined in Stage 3 apply to her alone. The woman's realised marriage-market payoff if the match is accepted is

$$U_j^f(a) = V_j(a+1) + \theta\,\Delta_j(a), \tag{7}$$

where $\theta \in (0,1)$ is a reduced-form retention parameter: the fraction of her gross marriage gain $\Delta_j(a)$ that she is able to realise once intra-household bargaining frictions are accounted for. Since $\Delta_j(a)$ is defined as the woman's own private net gain from the match rather than a jointly generated surplus over which she and a groom bargain, the complementary $(1-\theta)\Delta_j(a)$ is simply dissipated by those frictions; it is not a payment or transfer to any other party in the model. The same bargaining primitive also shows up in Stage 3 through the Pareto weight $\omega < 1$, so θ governs both her Stage 2 retention rate and the Stage 3 within-marriage distortion as a single primitive.

The per-period flow utility while unmarried, $u^u(a)$, is bounded and weakly decreasing in a. This captures the social and economic pressures that intensify with age in sub-Saharan Africa: stigma, declining match quality, and family pressure are all consistent with a falling $u^u(a)$. For $a < a_1$, the value of remaining unmarried under jurisdiction j satisfies

$$V_j(a) = u^u(a) + \delta\Big[(1-\lambda(a))V_j(a+1) + \lambda(a)\max\{U_j^f(a), V_j(a+1)\}\Big], \tag{8}$$

with terminal condition $V_j(a_1) = \bar{V}$, common across jurisdictions under Assumption 1. The marriage hazard is $h_j(a) = \lambda(a)\mathbf{1}\{\Delta_j(a) \geq 0\}$. The discount factor δ in the marriage-market Bellman and the discount factor β in the household objective are distinct: δ discounts across marriage-market ages, while β discounts between period 1 and period 2 within the household.

Proposition 2 (Stricter enforcement reduces underage marriage hazards)**.** *For $a < \bar{a}_{Strict}$, if $\Lambda_{Strict}(a) \geq \Lambda_{Lenient}(a)$, then $h_{Strict}(a) \leq h_{Lenient}(a)$, with strict inequality whenever $\Delta_{Lenient}(a) \geq 0 > \Delta_{Strict}(a)$, i.e. whenever the wedge causes a woman to decline a match she would otherwise have accepted. This result holds in partial equilibrium, for a fixed continuation value, and extends to general equilibrium by backward induction from the terminal condition $V_j(a_1) = \bar{V}$; (Proof in Appendix A)*

The enforcement wedge provides the direct mechanism through which the strict regime shifts the distribution of marriage ages upward. A richer model with individual heterogeneity and reputation, in the spirit of Wahhaj (2018), would also generate selection effects above the legal threshold as women retained by enforcement differ from those who married earlier. We abstract from that channel and focus on the direct timing effect of the enforcement wedge.

D. Stage 3: The Married-Household Problem

Conditional on marriage at age a_m, the couple enters a two-period household problem with initial state

$$S_0(a_m, \xi) = \big(b_0(a_m, \xi), h_0(a_m, \xi), e_0(T_m; \kappa_j^*), n_0 = 0\big), \tag{9}$$

where b_0 denotes assets accumulated by the family on the daughter's behalf before marriage, h_0 female human capital, $e_0 = e_b + \kappa_j^* T_m^{\gamma}$ educational attainment, and ξ an i.i.d. match-specific draw on a compact support capturing idiosyncratic variation in assets, human capital, and match quality at the time of marriage, observed by both parties upon meeting but not before. The Pareto weight $\omega \in (0,1)$ is fixed within marriage and determined at entry by $\omega = \Omega(b_0, h_0, e_0; \theta)$, with θ the bargaining primitive introduced in the Setup. The mapping $\Omega(\cdot;\theta)$ is weakly increasing in each of (b_0, h_0, e_0) and weakly increasing in θ, so women who marry later, with higher entry states, enter marriage with greater bargaining power; the micro-foundation for this monotonicity is discussed in Appendix A.

In period 1, the household allocates the woman's unit time endowment among birth (f_1), continued schooling (s_1), and labour-market participation (ℓ_1), with $f_1 + s_1 + \ell_1 = 1$ and each indicator binary, so exactly one activity is chosen. The household maximises

$$W(S_0, \omega) = \max_{f_1, s_1, \ell_1} \omega\big[\log c_1 + \upsilon_f f_1 - \chi_s s_1 - \chi_\ell \ell_1\big] + (1-\omega)\big[\log c_1 + \upsilon_m f_1\big] + \beta V_2(S_1), \tag{10}$$

subject to $c_1 \le w^f(h_0, e_0)\ell_1 + y_m + Rb_0$, where $y_m \ge 0$ is the man's exogenous income and $R > 0$ is the gross return on assets. Parameters are: $\upsilon_f \ge 0$ and $\upsilon_m > \upsilon_f$, the woman's and man's utility benefits from a child; $\chi_s > 0$, the woman's non-pecuniary cost of continued schooling; $\chi_\ell > 0$, her disutility of labour; $\phi_s, \phi_\ell > 0$, human capital accumulation rates from schooling and work; and $w^f(h_0, e_0)$, the female wage, strictly increasing in both arguments. Because the time allocation is binary, only one activity is chosen, and when $\ell_1 = 0$ consumption equals $y_m + Rb_0$. We assume throughout that $y_m + Rb_0 > 0$, so that log consumption is well defined. Period-1 choices update the state: $h_1 = h_0 + \phi_\ell \ell_1 + \phi_s s_1$,

$e_1 = e_0 + s_1$, and $n_1 = f_1$. The net payoffs from each activity are

$$\begin{aligned}
\Pi_f &= \log(y_m + Rb_0) + \omega \upsilon_f + (1-\omega)\upsilon_m + \beta V_2(h_0, e_0, 1), \\
\Pi_s &= \log(y_m + Rb_0) - \omega \chi_s + \beta V_2(h_0 + \phi_s, e_0 + 1, 0), \\
\Pi_\ell &= \log\big(w^f(h_0, e_0) + y_m + Rb_0\big) - \omega \chi_\ell + \beta V_2(h_0 + \phi_\ell, e_0, 0).
\end{aligned}$$

The household selects $\max\{\Pi_f, \Pi_s, \Pi_\ell\}$.

Assumption 5 (Child-utility gap exceeds period-1 costs)**.** $\upsilon_m - \upsilon_f > \max\{\chi_s, \chi_\ell\}$.

Assumption 6 (Period-2 continuation value)**.** *The period-2 value function $V_2(h_1, e_1, n_1)$ is the joint household continuation value. It is bounded, strictly increasing in (h_1, e_1), and satisfies $V_2(h_1, e_1, 1) < V_2(h_1, e_1, 0)$ for all (h_1, e_1). The gain from avoiding a child is increasing in both arguments, uniformly over all (h_1, e_1) and all $\omega \in (0,1)$. Additional cross-point conditions required for Lemmas 1 and 2 are stated in Appendix A.*

Lemma 1 (Education raises the return to schooling relative to birth)**.** *Under Assumption 6, $\Pi_s - \Pi_f$ is strictly increasing in e_0. There exists a threshold $\bar{e}(b_0, h_0; \theta)$, implicitly defined by $\Pi_s = \Pi_f$ (with $\omega = \Omega(b_0, h_0, \bar{e}; \theta)$ satisfied jointly), such that schooling is preferred to birth whenever $e_0 > \bar{e}(b_0, h_0; \theta)$. The dependence on h_0 runs through the continuation V_2 (which enters Π_s and Π_f at different state arguments) and through ω; the dependence on b_0 runs through ω alone, since the consumption term $\log(y_m + Rb_0)$ is common to Π_s and Π_f and cancels in their difference. (Proof in Appendix A)*

Lemma 2 (Human capital raises the return to work relative to birth)**.** *Under Assumption 6, $\Pi_\ell - \Pi_f$ is strictly increasing in h_0. There exists a threshold $\bar{h}(b_0, e_0; \theta)$, implicitly defined by $\Pi_\ell = \Pi_f$, such that work is preferred to birth whenever $h_0 > \bar{h}(b_0, e_0; \theta)$. The dependence on e_0 runs through V_2 and through the wage $w^f(h_0, e_0)$; the dependence on b_0 runs through ω and through the wage gain $\log(w^f + y_m + Rb_0) - \log(y_m + Rb_0)$, which is decreasing in b_0. (Proof in Appendix A)*

Lemma 3 (Higher entry states delay fertility and shift the activity choice)**.** *Under Lemmas 1 and 2:*

(i) $f_1 = 0$ *whenever* $e_0 > \bar{e}(b_0, h_0; \theta)$ *or* $h_0 > \bar{h}(b_0, e_0; \theta)$*;*
(ii) $s_1 = 1$ *is optimal when* $e_0 > \bar{e}(\cdot)$ *and* $\Pi_s \geq \Pi_\ell$*;*
(iii) $\ell_1 = 1$ *is optimal when* $h_0 > \bar{h}(\cdot)$ *and* $\Pi_\ell \geq \Pi_s$*.*

The period-1 birth probability is weakly decreasing in (e_0, h_0). Above the thresholds, schooling is weakly increasing in e_0 (when $\Pi_s \geq \Pi_\ell$) and labour participation is weakly

increasing in h_0 (when $\Pi_\ell \geq \Pi_s$). Below the thresholds, the allocation among $\{f_1, s_1, \ell_1\}$ is determined by the relative magnitudes of $\{\Pi_f, \Pi_s, \Pi_\ell\}$, and the comparative static on any single activity with respect to (e_0, h_0) is not signed without further restrictions. (Proof in Appendix A)

Lemma 4 (Lower bargaining power weakly raises the birth rate)**.** *Under Assumption 5, $\Pi_f - \Pi_s$ and $\Pi_f - \Pi_\ell$ are both strictly decreasing in ω. A lower bargaining weight therefore expands the parameter region in which birth is chosen. (Proof in Appendix A)*

Remark 2. When ω is low because the woman entered marriage young with low (e_0, h_0), her preferences carry less weight and the household tilts toward the man's preferred outcome of more children. The same channel reinforces the activity-mix shift away from schooling and work, which are the inputs into human capital required for skilled occupations later.

The woman's outside option from leaving marriage at the end of period 1 is approximated by $\bar{V}^f(e_1, h_1, n_1) = \psi_e e_1 + \psi_h h_1 - \psi_n n_1$, with $\psi_e, \psi_h, \psi_n > 0$. This is a reduced-form approximation to the equilibrium re-entry value; the linear specification preserves the qualitative comparative statics on (e_1, h_1, n_1) that drive the prediction on exits from marriage. Children reduce the outside option because they lower remarriage prospects. She exits if $\bar{V}^f(e_1, h_1, n_1) > \omega V_2(h_1, e_1, n_1)$, where ωV_2 is her share of the household continuation value from remaining in the marriage. By Assumption 4, the man's participation does not bind, so exit is a one-sided decision.

Remark 3 (Entry states and the outside-option channel). When $f_1 = 1$ is chosen (in the region $e_0 \leq \bar{e}(\cdot)$ and $h_0 \leq \bar{h}(\cdot)$), no human capital or education accumulates ($h_1 = h_0$, $e_1 = e_0$) and children lower the outside option by $\psi_n > 0$. Early marriage therefore weakens the outside option through lower (e_1, h_1) and higher n_1. Whether this translates into a lower observed rate of exits depends on the relative magnitude of the corresponding shift in the stay value ωV_2.

The monotonicity of e_0 in a_m follows directly from the accumulation technology in equation (2) and holds for any $\kappa_j^* > 0$ and $\gamma > 1$; it is a consequence of the model rather than a maintained assumption.

Assumption 7 (Monotone marital entry states)**.** *The mappings $b_0(a, \xi)$ and $h_0(a, \xi)$ are weakly increasing in the age at marriage a. Combined with the strict monotonicity of e_0 in a_m noted above, all three entry-state components are weakly increasing in a_m.*

Assumption 8 (Dominance of asset and human-capital effects below the thresholds)**.** *In*

the region where $e_0 \leq \bar{e}(\cdot)$ and $h_0 \leq \bar{h}(\cdot)$, the positive effects of higher (b_0, h_0, e_0) on Π_f through consumption and through V_2 jointly dominate the negative effect operating through the higher Pareto weight $\omega = \Omega(b_0, h_0, e_0; \theta)$, so the net effect of an increase in a_m on $M(a_m)$, evaluated at the equilibrium choice $f_1 = 1$, is non-negative.

Proposition 3 (Marriage age and the continuation value from marriage)**.** *Under Assumptions 7 and 8 and Lemmas 1–3, the woman's value from entering marriage is weakly increasing in age at marriage:*

$$a' \geq a \implies M(a'; \kappa_j^*) \geq M(a; \kappa_j^*).$$

(Proof in Appendix A)

Remark 4. If marrying later raises the woman's marital entry state, why does early marriage arise in equilibrium? The answer is that $M(a)$ is the value conditional on marrying at age a, not the value of waiting. A woman who declines a match at age a does not receive $M(a+1)$ with certainty. Instead, she enters the next period unmarried, receives the continuation value $V^j(a+1)$, faces a weakly lower probability of meeting another potential groom, $\lambda(a+1) \leq \lambda(a)$, and may also face lower unmarried flow utility because $u^u(a)$ is weakly decreasing in age. Early marriage can therefore arise even when later marriage would generate a better marital entry state, because waiting is risky and the value of remaining unmarried may decline with age.

Proposition 4 (Integrated downstream implications of delayed marriage)**.** *Suppose the strict regime induces a weak upward shift in the distribution of age at marriage. Under Assumption 7 and Lemmas 1–4:*

- *(a) the distributions of e_0, h_0, and b_0 at marital entry shift weakly upward, and consequently e_1 and h_1 shift weakly upward;*
- *(b) the distribution of age at first birth shifts weakly upward;*
- *(c) the period-1 birth rate shifts weakly downward; labour-market choices with higher human-capital returns become more likely, mapping to high-skill occupations in the labour market; broad labour-force participation is ambiguous because fertility at low entry states competes with work;*
- *(d) the distribution of ω shifts weakly upward, weakly reducing the period-1 birth rate through Lemma 4;*
- *(e) the outside option $\bar{V}^f(e_1, h_1, n_1)$ shifts weakly upward, because delayed marriage yields higher (e_1, h_1) and lower n_1. The stay value $\omega V_2(h_1, e_1, n_1)$ also shifts weakly*

upward, since both ω and V_2 rise with delayed marriage. Whether this translates into more observed exits depends on the relative magnitudes of the two shifts and is therefore theoretically ambiguous.

(Proof in Appendix A)

E. Testable Predictions

Proposition 4 maps the model's recursive structure onto a set of observable cross-border predictions. Some follow directly from the direction of the comparative statics; others concern the pattern of effects across outcomes, which carries stronger diagnostic value.

Education: Propositions 1 and 4(a) jointly imply that women under stricter legal regimes accumulate more education. Proposition 1 establishes the pre-market channel: families invest more before the marriage market opens, because the expected return rises with anticipated marriage age. Stage 3 adds the within-marriage channel: women who marry later face less truncation of schooling already in progress. The pre-market channel also implies that the effect should be visible at the school-entry margin, since families who expect early marriage may withdraw their daughters from school before any marriage occurs.

Fertility timing and level: Proposition 4(b) predicts that age at first birth shifts upward under stricter enforcement, as delayed marriage delays the onset of childbearing. Proposition 4(c) and Lemma 4 predict lower completed fertility through two channels: women who enter marriage with higher education and human capital choose birth less often in period 1, and their higher Pareto weight shifts household decisions away from the man's preferred fertility outcome.

Skilled employment: Proposition 4(c) implies that skilled occupations should be more strongly affected than broad labour-force participation, which Lemma 3 leaves ambiguous below the thresholds. The mechanism is that the switch from birth toward schooling or work requires entry states to exceed $\bar{e}(\cdot)$ or $\bar{h}(\cdot)$, and the resulting human capital accumulation raises the returns to occupations where education has a steep gradient.

Urban and rural heterogeneity: The female wage function $w^f(h_0, e_0)$ has a steeper gradient in urban labour markets, where the returns to education are higher. On the lenient side, families forgo larger returns by underinvesting, so the cost of early marriage is higher in cities. The gain from switching to a strict regime is correspondingly larger in urban areas, operating through both the direct timing channel and the anticipatory investment channel. The model therefore predicts a wider strict-minus-lenient gap in education and

skilled employment in urban areas than in rural areas, providing a within-sample test of the mechanism.

Exits from marriage: Delayed marriage raises both the outside option $\bar{V}^f$ (through higher (e_1, h_1) and lower n_1) and the within-marriage stay value ωV_2 (through higher ω and higher V_2). The net effect on observed union dissolutions is therefore theoretically ambiguous, as Proposition 4(e) states.

IV. Data

A. Demographic and Health Surveys

The individual-level data come from the Demographic and Health Surveys (DHS), a programme of nationally representative household surveys conducted across sub-Saharan Africa continuously since the early 1990s. Each survey round interviews women aged 15–49, collecting detailed information on marriage, fertility, education, employment, and household decision-making. The DHS is the primary source of comparable individual-level data on women's outcomes across low-income countries. A key feature of the DHS design is that respondents are assigned GPS coordinates at the cluster level, which allows us to locate each woman relative to international borders. We use these coordinates to assign each woman to a border pair and to compute her signed distance from the relevant legal cutoff.

B. World Policy Analysis Center

Legal minimum marriage ages come from the World Policy Analysis Center (WORLD), which tracks national family law across countries and years. We use WORLD to construct a panel of minimum legal marriage ages with parental consent for all sub-Saharan African countries from 1990 onwards. For each potential border pair, we verify the minimum age on each side and identify the years during which a legal age gap exists. A pair is included in the analysis only if the legal age difference has been in force for at least 20 years, ensuring that a sufficiently large cohort of women has been exposed to each regime. We restrict to pairs where one side has a minimum age of 18 or above and the other has a minimum age strictly below 18, creating a binary contrast between strict and lenient jurisdictions. The twenty-year threshold follows from the width of the contested age window. For most border pairs, that window spans only two or three years, and women pass through it quickly. A shorter exposure period concentrates the estimation on one or two thin birth cohorts, making results

sensitive to survey timing rather than the legal environment. Twenty years typically spans multiple DHS rounds, giving the sample women from several distinct cohorts who were exposed to the legal regime at the relevant ages.

The 12 border pairs used in the analysis, along with their legal ages and exposure periods, are listed in Table 6 in the appendix. Figure 1 shows the geographic distribution of treatment and control countries across sub-Saharan Africa.

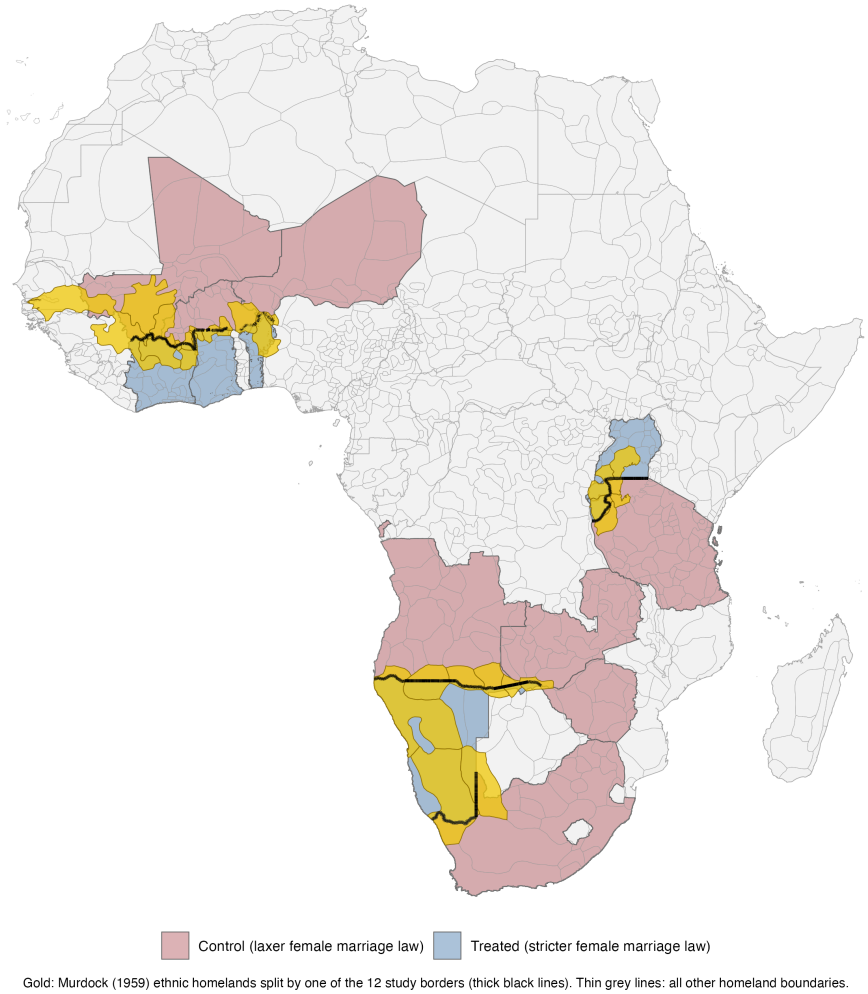


Figure 1: Treatment and Control Countries

C. Pixel-Level Geographic Controls

To absorb fine-grained variation in local development conditions near the border, we construct a set of pixel-level geographic controls matched to each DHS cluster using GPS coordinates: average light density, population density, elevation, travel time to the nearest

market, rainfall, temperature, distance to the sea coast, land suitability for agriculture, the malaria suitability index, distance to the capital city, and area under water (rivers, lakes, streams, and wetlands). These variables are drawn from rasters of differing native resolution (from 30 arc-seconds for light density up to 0.5° for several others) and, where relevant, assigned via a 12.5 km × 12.5 km grid; full sourcing and construction details, including the resolution of each variable, are documented in Appendix A.2.

A further set of geographic covariates was also collected and is reported in the covariate balance table (Table 5) in Appendix A.3 rather than included in the main control vector: indicators for a diamond mine and a petroleum/oil field in the pixel, distance to historical Christian missions, distance to colonial-era railways, distance to pre-colonial explorer trade routes, and country-level GDP per capita and terrain ruggedness. These are excluded from the regression controls because, with only twelve border pairs, the design does not support separating the marriage-law discontinuity from other country- or pixel-level attributes with too little independent variation to identify separately (Appendix A.2).

D. Ethnicity Data

Ethnicity fixed effects are constructed using the Murdock (1959) map of pre-colonial African ethnic homelands, the standard source for assigning ethnicity to geo-referenced observations in the African economics literature. Because many ethnic homelands straddle international borders, the Murdock boundaries are largely orthogonal to the national legal boundaries that generate our treatment variation. We assign each DHS cluster to an ethnic homeland by overlaying the GPS coordinates of the cluster on the Murdock shapefile, and include a fixed effect for each ethnic group in all specifications. This absorbs time-invariant cultural, linguistic, and institutional variation at the group level, including customary marriage practices and kinship structures.

V. Empirical Strategy

A. Identification

The paper estimates a regression discontinuity design in which treatment assignment is determined by residence relative to an international border. Let RD_{ibet} denote the signed distance to the border cutoff for individual i in border pair b, ethnicity e, and survey year t. Positive values indicate residence on the treated side of the border. The estimation sample is

restricted to individuals within a symmetric bandwidth around the cutoff ($|RD_{ibet}| \leq 100$ km in the main specification).

The estimand is the total causal effect of exposure to a strict child marriage legal environment on women's outcomes. As established in Section III, the enforcement cost of underage marriage is the only primitive that differs across the border, and it operates through two simultaneous channels (realised marriage timing and anticipatory pre-market investment, Stages 2–3 and 1 respectively). Because the anticipatory channel acts even on families whose daughters' realised child-marriage status would not be shifted by the law, we do not treat residence on the high-age side as an instrument for realised child marriage: rescaling the reduced-form effects by a first-stage compliance rate would misattribute the anticipatory response to measurement noise rather than treating it as a real channel. We instead report the intent-to-treat (reduced-form) effect of legal exposure directly for every outcome, and we test whether legal exposure moves the incidence of child marriage itself using the same specification. That test tells us whether the legal discontinuity is a real, behaviourally binding margin, which is the premise the rest of the design relies on, without using the child-marriage coefficient to rescale any other estimate.

Let $f_b(RD_{ibet})$ denote a border-pair-specific slope control that allows the relationship between distance and outcomes to differ on each side of the cutoff within each border pair:

$$f_b(RD_{ibet}) = \alpha_b RD_{ibet} + \beta_b RD_{ibet} \cdot Treated_{ibet}, \tag{11}$$

where α_b is the slope on the control side and $\alpha_b + \beta_b$ is the slope on the treated side. This allows the running variable to kink at the cutoff, separately for each border pair.

Reduced form

$$Y_{ibet} = \alpha + \rho\, Treated_{ibet} + f_b(RD_{ibet}) + X'_{ibet}\gamma + \lambda_b + \delta_e + \tau_t + u_{ibet}. \tag{12}$$

Standard errors are clustered at the survey-cluster level. We estimate this equation for every outcome Y_{ibet}, including the child-marriage indicator CM_{ibet} itself, which lets us test directly whether legal exposure shifts the incidence of marriage before age 18 near the border (reported alongside the other outcomes in Table 1).

B. Variable Definitions

The outcome variable Y_{ibet} covers five measures drawn from the DHS: an indicator for multiple unions, years of completed education, age at first birth,[1] total children ever born, and age at first sex.[2] We additionally examine education level, an ordered categorical variable coded as 0 (no education), 1 (primary), 2 (secondary), and 3 (higher education). We also examine several additional employment and behavioural outcomes. Not working is an indicator equal to one if the woman reports no current occupation, and zero otherwise. We also examine domestic-service employment. Domestic work is an indicator equal to one if a working woman reports domestic service as her primary occupation, and zero otherwise among working women.[3] We also construct two white-collar employment indicators. White-collar (extensive) equals one if the woman reports a professional or technical occupation, and zero otherwise. White-collar (intensive) is defined analogously but restricted to working women only.[4] We also examine child marriage directly: CM_{ibet} is an indicator equal to one if the individual married before age 18, and zero otherwise. The treatment indicator $Treated_{ibet}$ equals one for individuals residing on the high-age side of the border. The running variable RD_{ibet} is the signed distance from the border cutoff, with positive values indicating residence on the treated side. The slope control $f_b(RD_{ibet})$ is a border-pair-specific function of the running variable that allows the slope to differ on each side of the cutoff within each border pair, as specified above. The control vector X_{ibet} includes age, and an indicator for urban residence. The fixed effects λ_b, δ_e, and τ_t correspond to border pair, ethnicity, and survey year, respectively. Together they absorb time-invariant differences across border segments and ethnic groups, as well as aggregate shocks that are common across survey rounds.

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[1] Age at first birth is defined only for women who have given birth at least once; women who have not yet given birth are excluded from this outcome.

[2] Age at first sex uses DHS variable `v525`, retaining only numeric age responses. Responses coded as never had sex (0), at first union (96), inconsistent (97), do not know (98), or missing (99) are excluded. We do not recode the "at first union" response to age at first union in order to avoid mechanically linking this outcome to marriage timing. Because both outcomes are undefined for women who have not yet experienced the event, Appendix A.4 reports Lee (2009) bounds (Table 18) addressing the concern that legal exposure could itself alter who is observed to have reached these milestones by the survey date.

[3] Domestic work is defined only among working women since the question of occupational category is not meaningful for non-workers. The smaller observation counts for this outcome reflect this restriction.

[4] Working women with missing occupation codes are excluded from both white-collar outcomes rather than coded as zero, since their true occupational category is unknown. Coding them as non-white-collar would incorrectly classify them as low-skill workers and could bias the estimates downward.

C. Identifying Assumption

The first identifying assumption requires that residence on the high-age side of the border affects outcomes only through the child marriage legal environment, and not through any independent channel correlated with both treatment and the outcomes. Our design follows the border-institution logic of Anderson (2018), which imposes the additional assumption that potential outcomes would vary smoothly at the border absent the discontinuity in the relevant legal environment. We address the standard threats to these assumptions through several complementary arguments.

The most direct threat is that the two sides of a border differ in cultural practices, kinship systems, or social norms in ways that independently shape marriage behaviour and downstream outcomes. The ethnicity fixed effects δ_e (constructed as described in Section IV) directly address this concern: because Murdock ethnic homelands straddle national borders, these fixed effects compare women within the same pre-colonial ethnic group on opposite sides of the legal threshold, holding kinship structure, customary marriage practices, and community norms constant.

A second concern is that the high-age country may simply be richer or better-served by infrastructure, producing better outcomes for women through channels unrelated to marriage law. To address this, every specification includes eleven pixel-level controls matched to each DHS cluster using GPS coordinates: nighttime light intensity, population density, agricultural suitability, the malaria ecology index, distance to the national capital, water and wetland coverage, elevation, travel time to market, rainfall, temperature, and distance to coast (full sourcing and native resolutions in Appendix A.2). The main estimates are robust to the inclusion of these controls, which speaks against the development gradient as an alternative explanation. Border-pair fixed effects λ_b absorb remaining time-invariant differences at the border-segment level, and the border-pair-specific slopes $f_b(RD_{ibet})$ allow for smooth within-country gradients in conditions as a function of distance from the cutoff.

The migration robustness check reported in Appendix A.4 provides further support for the identification strategy: excluding individuals who migrated within three years before their marriage leaves the main estimates largely unchanged.

A potential concern is that minimum marriage age laws form part of a broader policy bundle, co-moving with compulsory schooling age, minimum employment age, and age of consent in ways the treatment indicator would capture in their entirety. We address this di-

rectly using data from the WORLD Policy Analysis Center on compulsory schooling ages and minimum employment ages for all countries in our sample. Neither policy dimension co-moves systematically with the minimum marriage age across our 12 border pairs: the direction of differences varies across pairs, with some pairs showing the control country stricter, others the treatment country stricter, and several pairs identical. We also assess the broader legal environment using the World Bank's Women, Business and the Law indicators across five dimensions, reported in Table 21. The pattern is broadly balanced and does not show a systematic advantage on the treatment side. Namibia warrants specific attention. The four Namibia border pairs are the only pairs where the treatment country has both a stricter marriage age law and a stricter compulsory schooling age; dropping these pairs meaningfully reduces the education-years effect (Table 12), so compulsory schooling should be read as a contributing channel alongside the marriage-age law rather than a fully ruled-out confound, though the education effect remains positive and significant outside the Namibia pairs. Niger, as a control country, has weaker laws across multiple dimensions, meaning the control side in the Benin–Niger pair is doubly disadvantaged relative to the treatment side; since Niger contributes only a single border pair, this concern is addressed directly by the leave-one-border-pair-out check (Table 16), which shows that dropping Benin–Niger alone moves every coefficient only modestly from its pooled value, confirming that this outlier does not drive the main estimates.

Other threats regard multiple hypothesis testing and spatially correlated errors. These are addressed via Romano-Wolf correction and Conley standard errors in Section VII. Table 4 in the appendix summarises these threats and responses.

Full covariate balance tests at the cutoff are reported in Table 5 in Appendix A.3, together with a discussion of which imbalances motivate the regression controls used throughout.

D. Sample and Border Pairs

The estimation sample is drawn from the DHS surveys described in Section IV, restricted to women within a symmetric bandwidth of the relevant border cutoff. The analysis covers twelve border pairs, listed in Table 6 in the appendix together with the minimum legal marriage age on each side and the period during which the cross-border legal difference is in force. Border pairs are included when four conditions are met: the two countries share a land border, their minimum legal marriage ages differ, at least one DHS survey round with GPS data is available for both sides of the border during the exposure period, and the legal

age difference has been in force for at least 20 years, ensuring a sufficiently large cohort of women exposed to each regime. Two pairs that satisfy the legal criterion were excluded: the Central African Republic–Cameroon pair and the Republic of Congo–Angola pair, both of which lacked sufficient DHS coverage on one side of the border.

One pair, Rwanda–Tanzania, warrants a brief note. Rwanda's minimum legal marriage age was 18 prior to 2016, the same threshold as the majority of pairs in the sample. Following a 2016 reform, the minimum age was raised to 21. Despite this, the definition of child marriage is held constant throughout the analysis: a woman is classified as having married as a child if and only if she married before age 18, regardless of the legal threshold in her country of residence. For the Rwanda–Tanzania pair, we additionally restrict identification to cohorts of women who were aged 15–18 during the pre-reform period, when Rwanda's legal threshold was 18 and directly comparable to the other pairs. This ensures that the identifying variation for Rwanda–Tanzania comes from the same margin as the rest of the sample (the 18-year threshold) and is not contaminated by the post-reform discontinuity at 21.

The main specification uses a 100 km bandwidth around the cutoff, with 50 and 200 km serving as alternative specifications. Data-driven bandwidths were not used as these would yield different distances and polynomials for each outcome variable, altering the samples for each and would not ensure comparability. We exclude women aged under 18 with missing age at first marriage, as these observations cannot be used to construct the child marriage indicator. This restriction affects a small number of young never-married women and does not materially alter the sample composition.

E. Cohort Analysis

For each border pair, we retain only women who were within the early marriage-age window, defined as between the control country's minimum legal age and 18, during the years in which the cross-border legal difference was in force. For example, if the legal age gap between a pair operated from 1995 to 2023, and the relevant age window is 16–18, we include only women who were aged 16–18 throughout 1995–2023. This restriction means that every woman in the estimation sample was directly exposed to the legal regime during the period when it was binding for her. Unlike designs that include women of all ages and rely on cohort variation to identify exposure, our sample is already restricted to the most plausible complier population by construction: women for whom the minimum legal age was a potentially binding constraint during their early marriage years. This makes the

estimates more precisely targeted but also means cohort variation cannot be used as an additional source of identifying variation within the sample. Summary statistics for the full country samples appear in Tables 7 and 8 in the appendix.

VI. Results

A. Main Estimates

Table 1 reports the reduced-form estimates across all outcomes and bandwidth choices. Each coefficient measures the intent-to-treat effect of residing on the strict side of the border on the outcome. The main specification uses a 100 km bandwidth; the 50 and 200 km results serve as alternative specifications.

Table 1: Reduced-Form Effect of Treatment Across Bandwidths

	Bandwidth (km)		
Outcome	50	100	200
Marriage			
Married before age 18 (child marriage)	−0.141***	−0.172***	−0.159***
	(0.022)	(0.018)	(0.015)
Obs.	27,508	59,436	103,685
Human capital			
Education (years)	0.962***	1.205***	1.298***
	(0.227)	(0.195)	(0.179)
Obs.	27,508	59,436	103,685
Education level	0.282***	0.334***	0.334***
	(0.040)	(0.035)	(0.032)
Obs.	27,521	59,461	103,716
Fertility			
Age at first birth	0.857***	0.931***	0.755***
	(0.154)	(0.123)	(0.104)
Obs.	19,680	40,857	71,803
Children ever born	−0.266***	−0.330***	−0.280***
	(0.070)	(0.055)	(0.047)
Obs.	27,521	59,461	103,720
Age at first sex	0.527***	0.633***	0.480***
	(0.124)	(0.109)	(0.091)
Obs.	19,236	39,717	70,981
Household and labour market			
Multiple unions	0.010	−0.008	−0.011
	(0.011)	(0.008)	(0.007)
Obs.	27,511	59,444	103,681
Not working	0.103***	0.113***	0.149***
	(0.028)	(0.022)	(0.019)
Obs.	27,488	59,404	103,632
Domestic work	0.005	−0.016***	−0.008*
	(0.005)	(0.005)	(0.004)
Obs.	18,419	39,807	68,120
White-collar (extensive)	0.026***	0.029***	0.026***
	(0.009)	(0.007)	(0.007)
Obs.	27,122	58,588	102,098
White-collar (intensive)	0.033***	0.044***	0.044***
	(0.012)	(0.010)	(0.010)
Obs.	18,419	39,807	68,120

Notes: Reduced-form estimates. The treatment indicator equals one for women residing on the high-age side of the border. Standard errors clustered at the DHS survey-cluster level in parentheses. All regressions include border-pair fixed effects, border-pair-specific slopes and kinks, pixel-level geographic controls, ethnicity fixed effects, and survey-year fixed effects. Because a small number of outcomes exclude some women (for example, age at first sex excludes women who have never had sex), the exact child-marriage coefficient varies marginally across outcome-specific estimation samples; the value reported here is the one common to most samples (see Table 17 in Appendix A.4 for the coefficient estimated separately on each outcome-specific sample). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Marriage: Before turning to the downstream outcomes, we test the mechanism the rest of the design relies on: does legal exposure actually move the child marriage rate? Residing on the strict side lowers the probability of marrying before age 18 by roughly 14 to 17 percentage points, stable across bandwidths and significant at the 1 percent level. This confirms that the legal discontinuity is a real behavioural margin near the border rather than a distinction that exists on paper only. We use this coefficient only as evidence that the mechanism is active; we do not use it to rescale the other reduced-form effects below, for the reasons discussed in Section V.

Human capital: The education results operate through two channels, both introduced in the model and pulling in the same direction. The first is within-marriage truncation: marriage can truncate formal schooling, removing years on the steepest part of the accumulation curve. The second is pre-market underinvestment: families on the lenient side of the border, anticipating early marriage, invest less in daughters' education before the marriage market is ever reached. Figure 2 illustrates the two channels. Residing on the strict side raises educational attainment by 0.33 grade levels at the 100 km bandwidth, significant at the 1 percent level and stable across bandwidths and sample-composition checks; we treat this as the primary human-capital result. The continuous years measure also rises, by 1.21 years, but is considerably less stable, losing significance at 50 km without Namibia (Table 12) and falling to an insignificant 0.120 without Ghana–Burkina Faso (Table 16).

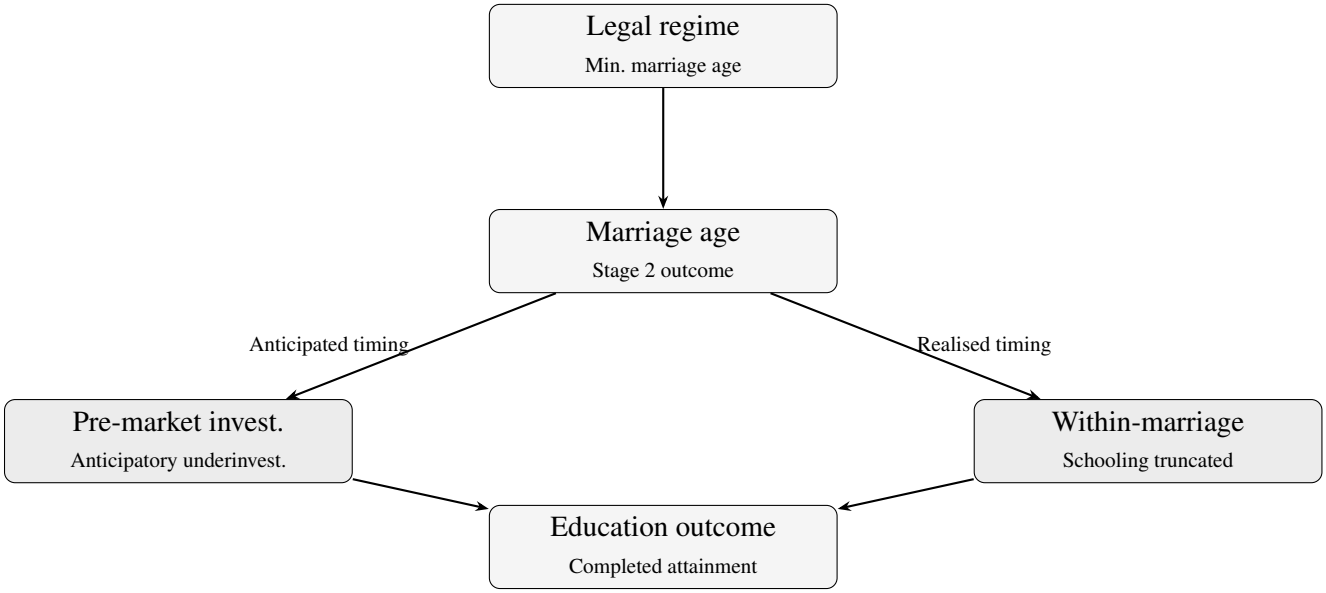


Figure 2: Two channels from the legal environment to education

Table 2 decomposes the education effect into extensive and intensive margins.[5] Residing on the strict side raises the probability of receiving any education by 16.5 percentage points at the 100 km bandwidth, significant at the 1 percent level and stable across bandwidths. Conditional on some schooling, the strict side also adds 0.444 further years at 100 km, significant at the 5 percent level and rising to 0.647 years, also significant at the 1 percent level, at 200 km. Both margins therefore contribute to the overall effect: the school-entry margin still accounts for a sizable share of the increase in receiving any education, but the additional years of schooling conditional on enrollment are also economically meaningful and statistically significant.

Table 2: Reduced-Form Effects: Education Margins

	Bandwidth (km)		
Outcome	50	100	200
Education (years)	0.962^{***}	1.205^{***}	1.298^{***}
	(0.227)	(0.195)	(0.179)
Any education	0.122^{***}	0.161^{***}	0.158^{***}
	(0.023)	(0.020)	(0.018)
Education (years $\mid$ schooling > 0)	0.282	0.444^{**}	0.647^{***}
	(0.227)	(0.189)	(0.165)
Education level	0.282^{***}	0.334^{***}	0.334^{***}
	(0.040)	(0.035)	(0.032)

Notes: Reduced-form estimates. The treatment indicator equals one for women residing on the high-age side of the border. Standard errors clustered at the DHS survey-cluster level in parentheses. All specifications include border-pair fixed effects, border-pair-specific slopes and kinks, pixel-level geographic controls, ethnicity fixed effects, and survey-year fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Fertility: Women on the strict side of the border have their first birth 0.93 years later and have 0.33 fewer children ever born, both at the 100 km bandwidth. These patterns are consistent with Stage 3 of the model: women who enter marriage with lower human capital and a weaker bargaining position face earlier and higher fertility. The age-at-first-birth effect is stable across all bandwidths, which is expected given the close link between marriage timing and onset of childbearing in this context.

Age at first sex is 0.63 years later on the strict side of the border at the 100 km bandwidth, significant at the 1 percent level. The model does not formally predict this outcome, but the direction is consistent with marriage timing as a driver of sexual debut.

[5]The margin decomposition uses the same reduced-form specification as Table 1.

The age at first sex variable is non-missing only for women who have had sex, so the sample drops women who had not yet had sex, a group likely correlated with treatment on the strict side. This sample restriction also shrinks the effective sample used to estimate the effect of legal exposure on child marriage for this outcome, relative to the sample used for the other outcomes. The reduced-form estimate should be read as the effect among sexually active women rather than the full population, which is why we treat it as auxiliary evidence rather than a core prediction.

Household and labour market: Multiple-union status shows no significant effect across any bandwidth. The model's prediction here is about exit from marriage, which Proposition 4(e) leaves theoretically ambiguous: delayed marriage raises both the woman's outside option $\bar{V}^f$ (through higher (e_1, h_1) and lower n_1) and her within-marriage stay value ωV_2 (through higher ω and higher V_2), and which shift dominates is not pinned down by the model. Multiple-union status is in any case an imperfect proxy for exit: it captures voluntary dissolution but also widowhood, polygamy, abandonment, and remarriage propensity, each of which may vary across our twelve border pairs and is not held constant by the legal marriage-age discontinuity.

On employment, residing on the strict side raises the not-working rate by 11.4 percentage points at the 100 km bandwidth ($p < 0.01$), rising to 14.7 points ($p < 0.01$) at 200 km. This implies that a stricter legal environment raises the share of women who are not working. The model is largely silent on broad labour-force participation as a clean prediction (Lemma 3 leaves it ambiguous because fertility at low entry states competes with work). The fact that the coefficient runs counter to a simple "bargaining raises participation" intuition indicates that the broad-participation outcome is not the right margin to test the model's mechanism. The model also abstracts from male income endogeneity, household consumption-smoothing, and selection into informal work, all of which could plausibly drive women on the lenient side into low-skill participation.

The white-collar results are more consistent with the model. Residing on the strict side raises the probability of white-collar employment by 2.9 percentage points on the extensive margin and by 4.9 percentage points among working women, both significant at the 1 percent level and stable across bandwidths. Because skilled occupations have steeper returns to education, they are the first margin affected by an accumulation loss. Domestic work is significant at the 100 km bandwidth only, with no significant effect at 50 or 200 km.

B. Heterogeneity by Urban and Rural Residence

Table 3 splits the reduced-form estimates by urban and rural residence and presents results at the 100 and 200 km bandwidths. The 50 km bandwidth is excluded for urban women because the urban subsample within 50 km of the border contains only 5,669 women, which is too small for reliable inference. The education effects are significant in both subsamples but substantially larger in urban areas. Residing on the strict side raises education level by 0.50 levels for urban women and 0.32 for rural women at the 100 km bandwidth, both significant at the 1 percent level. Educational attainment also rises by 2.7 years at 100 km for urban women, compared with 1.1 years in rural areas.

A note of caution is warranted in reading these results. The point estimate in education-years is somewhat larger at the 100 km bandwidth (2.7) than at 200 km (1.5) for urban women, though both are significant at the 1 percent level. The rural estimate is comparatively stable across bandwidths, at 1.1 years at 100 km and 1.2 years at 200 km. The 100 km urban result remains the largest single number in the heterogeneity table and should be read with some caution given the smaller urban subsample.

The urban-rural pattern is consistent with at least two distinct mechanisms. One is steeper returns to education in urban labour markets: when $\partial w^f / \partial e_0$ is larger, the truncation of the convex accumulation path κT_m^{γ} at an early age entails a greater loss in lifetime human capital, and families on the urban side rationally invest more in κ to begin with, amplifying the attainment gap. A second is the reach of statutory law: urban areas have better access to formal legal institutions, so the minimum legal marriage age is more binding there, and the discontinuity at the border is correspondingly sharper. In rural areas, customary norms and weak state capacity mean the formal law reaches fewer people. These two mechanisms are observationally equivalent in our design and we cannot distinguish them.

The same two mechanisms apply to the fertility and white-collar results. Urban women on the strict side reach age at first birth 1.68 years later than urban women on the lenient side, against 0.78 years for rural women, and the white-collar employment effect is substantially larger in urban areas, roughly four times larger on both the extensive and intensive margins. Both patterns are consistent with the model: stronger urban education effects push more women above the $\bar{e}$ and $\bar{h}$ thresholds in Lemmas 1 and 2, shifting period-1 choices away from birth and toward schooling or skilled work, and the steeper urban wage gradient amplifies the white-collar margin specifically. The urban-rural decomposition is therefore consistent across the chain from education to fertility to skilled employment.

Table 3: Reduced-Form Effects by Urban and Rural Residence

	Urban		Rural	
Outcome	100	200	100	200
Human capital				
Education (years)	2.678***	1.499***	1.074***	1.180***
	(0.480)	(0.453)	(0.208)	(0.194)
Education level	0.511***	0.343***	0.319***	0.321***
	(0.085)	(0.080)	(0.038)	(0.036)
Fertility				
Age at first birth	1.679***	1.239***	0.780***	0.644***
	(0.310)	(0.221)	(0.135)	(0.118)
Children ever born	−0.585***	−0.294***	−0.304***	−0.264***
	(0.106)	(0.087)	(0.061)	(0.055)
Age at first sex	1.342***	0.834***	0.550***	0.442***
	(0.259)	(0.208)	(0.123)	(0.103)
Household and labour market				
Multiple unions	−0.051***	−0.054***	0.001	−0.001
	(0.017)	(0.013)	(0.010)	(0.009)
Not working	0.077*	0.031	0.124***	0.147***
	(0.047)	(0.036)	(0.026)	(0.023)
Domestic work	−0.015	−0.017	−0.003	−0.000
	(0.022)	(0.015)	(0.002)	(0.002)
White-collar (extensive)	0.094***	0.074***	0.020***	0.017***
	(0.026)	(0.022)	(0.006)	(0.005)
White-collar (intensive)	0.166***	0.105***	0.030***	0.027***
	(0.039)	(0.033)	(0.008)	(0.007)

Notes: Urban and rural subsamples defined using DHS urban–rural classification. Standard errors clustered at the DHS survey-cluster level in parentheses. All specifications include border-pair fixed effects, border-pair-specific slopes and kinks, pixel-level geographic controls, ethnicity fixed effects, and survey-year fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

VII. Robustness Checks

We subject the main results to eight additional checks: excluding individuals who migrated shortly before marriage, adjusting inference for multiple hypothesis testing across outcome families using the Romano-Wolf procedure, using spatially robust (Conley) standard errors under a data-driven bandwidth, excluding the Namibia border pairs, which are the only pairs where the treatment country also has a stricter compulsory schooling age, excluding

the geographic and ethnicity controls entirely, re-estimating the main specification with 2nd- and 3rd-order border-pair-specific RD polynomials in place of the linear-with-kink polynomial, re-estimating the main specification while dropping one border pair at a time (including Niger's single pair) to check that no single pair drives the results, and computing Lee (2009) bounds for age at first birth and age at first sex, the two outcomes that are defined only on a treatment-affected subpopulation, to address the concern that legal exposure could itself alter who is observed to have reached these milestones by the survey date. Across five of the eight checks, the sign and significance pattern of the main results reported in Table 1 is preserved.[6] We report the full results and discussion of each check in Appendix A.4, alongside the balance tests and sample tables.

VIII. Discussion

The reduced-form estimates in Section VI show that women on the strict side of the border have more education, later first births, fewer children, later sexual debut, and greater access to skilled employment. The estimates are not equally informative for every outcome. The most stable findings, and the ones closest to the model's mechanisms, are educational attainment, age at first birth, children ever born, and white-collar employment. Broad labour-force participation also shifts significantly, but the model leaves its sign ambiguous and the empirical pattern is not easily mapped onto the bargaining channel. The evidence is weaker for domestic work and multiple unions, where the estimates are either small, statistically insignificant, or noisy across specifications.

A. Law, enforcement, and what the border design identifies

The model treats the legal regime as an enforcement wedge: the expected cost of contracting an underage marriage. The empirical design does not observe that wedge directly. It observes a discontinuity in the legal environment at the border and estimates the intent-to-

[6]The three exceptions are the Namibia-exclusion check, where the education-years coefficient falls by roughly half and loses significance at the 50 km bandwidth (education level remains significant throughout), the leave-one-border-pair-out check, where Ghana–Burkina Faso is a high-leverage pair and excluding it moves the continuous education-years coefficient out of significance (education level remains significant throughout), and the higher-order polynomial checks, where age at first sex weakens substantially in magnitude at the 100 km bandwidth (from 0.650^{***} to 0.363^{***}) though it remains significant at the 5 percent level throughout, and where a handful of outcomes lose significance at the 50 km or 200 km bandwidths under the 3rd-order polynomial (children ever born, domestic work, and white-collar extensive), even though the pattern at the 100 km bandwidth used for the main specification is stable across all eleven outcomes. See Appendix A.4 for the full results.

treat effect of that discontinuity directly, including on child marriage itself. The estimates show that stricter legal exposure changes marriage behaviour and lifetime outcomes near the border. They do not separately identify whether the effect comes from the statutory minimum age, actual enforcement, perceived enforcement, administrative barriers, or some combination of these.

The urban-rural heterogeneity is consistent with both a reach-of-law interpretation and a returns-to-schooling interpretation, and should motivate rather than establish the enforcement reading (Section VI. B).

The policy implication is therefore more limited than a simple enforcement claim. Legal environments with stricter minimum ages matter when they change the expected cost of underage marriage. That expected cost may come from courts, registration systems, sanctions, school reporting, community monitoring, or legal awareness. The paper does not estimate these channels separately. It shows that stricter legal environments are associated with lower child marriage near borders and with better outcomes across education, fertility, and skilled employment. The model explains why those effects may extend beyond marriage timing itself.

B. Limitations

Several limitations follow from the way the model and empirical design fit together.

The model separates pre-market investment, marriage-market timing, and household decisions after marriage. The empirical design estimates their combined effect. This is appropriate for the policy question, since a legal change can move all three margins at once. It is less informative about their relative importance. The education-margin decomposition supports the importance of school entry, but it does not observe parental expectations. The fertility estimates are consistent with a bargaining channel, but they do not separate bargaining power from the direct timing of marriage and childbirth.

The estimated effects apply to women whose marriage behaviour responds to the legal discontinuity: those who are likely close to the legal margin and whose outcomes may be especially sensitive to delayed marriage. The estimates should not be extrapolated to women whose marriages would occur early regardless of the law, or to settings where the legal threshold has no effect on behaviour.

The model gives bargaining power and outside options a central role, but the DHS outcomes are indirect. Fertility and white-collar employment are affected by bargaining in the

model, but they are not direct measures of bargaining power. Isolating these channels separately would require data on separation, divorce, control over income, assets, household decisions, fertility preferences, and outcomes after separation.

The theoretical framework abstracts from divorce, endogenous matching, post-marriage migration, child investment, spousal age gaps, and heterogeneity in male characteristics. These omissions are useful for tractability, but they narrow the interpretation. A stricter legal regime may affect whom women marry as well as when they marry. It may also affect transfers, match quality, and household formation in ways the current model does not separately identify.

The sample covers twelve border pairs in sub-Saharan Africa. The border design is valuable because it compares nearby populations exposed to different legal regimes, but the estimates remain local to the countries, cohorts, and institutions in the sample. The mechanisms may apply elsewhere, but the magnitudes need not. They will depend on schooling systems, labour markets, customary marriage rules, bride-price practices, state capacity, and norms around separation and remarriage.

C. Future research

A direct test of pre-marriage investment would be the clearest next step. The current evidence points toward the school-entry margin, but it cannot observe the family decision that the model places before marriage-market entry. A stronger test would follow girls before marriage and record enrolment, attendance, grade progression, parental expectations, and local marriage-market conditions. The central question is whether expected marriage timing changes education decisions before any marriage occurs.

Future work should also measure enforcement rather than infer it from legal exposure. The model's enforcement wedge could be proxied with local data on birth registration, marriage registration, court distance, legal-aid access, administrative sanctions, school reporting rules, or knowledge of the statutory age. These measures would help separate the law on paper from the institutions that make the law binding.

A third extension is to study bargaining with better outcomes. Multiple unions is too crude a proxy for exits to carry the outside-option interpretation. Future work should use direct measures of household decision-making, control over income, asset ownership, fertility preferences, separation, divorce, and post-separation welfare. These data would help distinguish improved bargaining within continuing marriages from increased ability to leave

marriages.

Another question is whether later interventions can offset some of the losses associated with early marriage. The model treats child marriage as producing a low entry state, but it does not imply that all later outcomes are fixed. Adult education, vocational training, childcare, contraception, legal aid, or cash transfers for already-married young women could raise human capital and outside options after marriage. The present paper cannot evaluate these policies, but they follow from the mechanism.

A structural extension would also be useful. A quantitative model with endogenous matching, spousal heterogeneity, bride-price transfers, divorce, and child investment could discipline counterfactual policy exercises. Such a model could compare statutory reform, enforcement, schooling subsidies, employment opportunities, and combined interventions. The value would be practical: to learn which margin explains the largest share of the reduced-form effects.

D. Policy implications

The results suggest three policy implications. First, minimum-age laws can matter when they change the effective environment in which marriage decisions are made. The evidence does not imply that raising the statutory age mechanically improves outcomes. Rather, it shows that women exposed to stricter legal regimes experience lower child-marriage rates and better outcomes in education, fertility timing, fertility levels, sexual debut, and white-collar employment. The relevant policy object is therefore not the legal text alone, but the extent to which the law changes the expected cost of underage marriage.

Second, legal reform is likely to be most effective when paired with enforcement and administrative capacity. In the model, the operative legal mechanism is the expected cost of underage marriage, which depends on whether households believe the rule is visible, credible, and costly to violate. In practice, this points to the importance of birth and marriage registration systems, local administrative capacity, access to legal information, and credible sanctions. A statutory threshold that is widely unknown or easily ignored is unlikely to produce the same effects as a legal regime that is actually enforced.

Third, child-marriage policy should be evaluated using a broader set of outcomes than marriage timing alone. The theoretical framework shows that marriage laws may affect families' decisions before marriage, women's bargaining position within marriage, and later-life outcomes after marriage. The empirical results are consistent with this broader

chain: the strict-law side shows improvements not only in child marriage itself, but also in schooling, fertility, and skilled employment. This means that policy evaluations focused only on the number of marriages delayed may understate the full effect of legal regimes.

IX. Conclusion

The main takeaway from this paper is that minimum marriage-age laws matter most when they alter the environment in which families and marriage markets make decisions. The evidence does not support a simple interpretation in which statutory reform mechanically transforms women's outcomes. Instead, the results point to a more specific mechanism: when legal regimes raise the expected cost of underage marriage, they can delay marriage, increase the returns to investing in girls' education, and improve the conditions under which women enter married households.

This interpretation helps connect the empirical findings. The effects on education, fertility timing, fertility levels, sexual debut, and white-collar employment are not separate outcomes moving independently. They are consistent with a single process in which delayed marriage changes both pre-marital investment and post-marital household decisions. The legal regime appears to move outcomes most clearly where education and human capital are central.

The findings therefore suggest that child-marriage law should be viewed as part of a broader human-capital and household-bargaining environment, not only as a narrow family-law intervention. At the same time, the results should be interpreted with caution. The empirical design identifies the effect of exposure to stricter legal regimes near international borders, not the universal effect of raising a statutory age in every institutional setting. The policy relevance of the findings depends on whether legal thresholds are credible, visible, and enforceable enough to change household expectations.

References

Siwan Anderson. Legal origins and female HIV. *American Economic Review*, 108(6): 1407–1439, 2018. doi: 10.1257/aer.20151047.

Nava Ashraf, Erica Field, and Jean Lee. Household bargaining and excess fertility: An experimental study in Zambia. *American Economic Review*, 104(7):2210–2237, 2014. doi: 10.1257/aer.104.7.2210.

Nava Ashraf, Natalie Bau, Nathan Nunn, and Alessandra Voena. Bride price and female education. *Journal of Political Economy*, 128(2):591–641, 2020. doi: 10.1086/704572.

Gary S. Becker. A theory of marriage: Part I. *Journal of Political Economy*, 81(4):813–846, 1973. doi: 10.1086/260084.

Gary S. Becker. A theory of marriage: Part II. *Journal of Political Economy*, 82(2):S11–S26, 1974. doi: 10.1086/260287.

Yoram Ben-Porath. The production of human capital and the life cycle of earnings. *Journal of Political Economy*, 75(4):352–365, 1967. doi: 10.1086/259291.

Martin Browning, Pierre-André Chiappori, and Yoram Weiss. *Economics of the Family*. Cambridge University Press, Cambridge, 2014. ISBN 9780521795395.

Nina Buchmann, Erica Field, Rachel Glennerster, Shahana Nazneen, and Xiao Yu Wang. A signal to end child marriage: Theory and experimental evidence from Bangladesh. *American Economic Review*, 113(10):2645–2688, 2023. doi: 10.1257/aer.20220720.

Pierre-André Chiappori. Rational household labor supply. *Econometrica*, 56(1):63–90, 1988. doi: 10.2307/1911842.

Pierre-André Chiappori. Collective labor supply and welfare. *Journal of Political Economy*, 100(3):437–467, 1992. doi: 10.1086/261825.

Pierre-André Chiappori, Bernard Fortin, and Guy Lacroix. Marriage market, divorce legislation, and household labor supply. *Journal of Political Economy*, 110(1):37–72, 2002. doi: 10.1086/324385.

Pierre-André Chiappori, Bernard Salanié, and Yoram Weiss. Partner choice, investment in children, and the marital college premium. *American Economic Review*, 107(8):2109–2167, 2017. doi: 10.1257/aer.20150154.

Matthew Collin and Theodore Talbot. Are age-of-marriage laws enforced? Evidence from developing countries. *Journal of Development Economics*, 160:102950, 2023. doi: 10.1016/j.jdeveco.2022.102950.

Lucia Corno, Nicole Hildebrandt, and Alessandra Voena. Age of marriage, weather shocks, and the direction of marriage payments. *Econometrica*, 88(3):879–915, 2020. doi: 10.3982/ECTA15505.

Flavio Cunha and James J. Heckman. The technology of skill formation. *American Economic Review*, 97(2):31–47, 2007. doi: 10.1257/aer.97.2.31.

Gordon B. Dahl. Early teen marriage and future poverty. *Demography*, 47(3):689–718, 2010. doi: 10.1353/dem.0.0120.

Esther Duflo, Pascaline Dupas, and Michael Kremer. Education, HIV, and early fertility: Experimental evidence from Kenya. *American Economic Review*, 105(9):2757–2797, 2015. doi: 10.1257/aer.20121607.

Aaron S. Edlin and Chris Shannon. Strict monotonicity in comparative statics. *Journal of Economic Theory*, 81(1):201–219, 1998.

Raquel Fernández and Joyce Cheng Wong. Divorce risk, wages and working wives: A quantitative life-cycle analysis of female labour force participation. *The Economic Journal*, 124(576):319–358, 2014. doi: 10.1111/ecoj.12136.

Erica Field and Attila Ambrus. Early marriage, age of menarche, and female schooling attainment in Bangladesh. *Journal of Political Economy*, 116(5):881–930, 2008. doi: 10.1086/593333.

Jorge Garcia-Hombrados. Child marriage and infant mortality: Causal evidence from Ethiopia. *Journal of Population Economics*, 35(3):1163–1223, 2022. doi: 10.1007/s00148-021-00873-y.

Rachel Heath and A. Mushfiq Mobarak. Manufacturing growth and the lives of Bangladeshi women. *Journal of Development Economics*, 115:1–15, 2015. doi: 10.1016/j.jdeveco.2015.01.006.

Guido W. Imbens and Thomas Lemieux. Regression discontinuity designs: A guide to practice. *Journal of Econometrics*, 142(2):615–635, 2008. doi: 10.1016/j.jeconom.2007.05.001.

Murat Iyigun and Jeanne Lafortune. Why wait? a century of education, marriage timing and gender roles. IZA Discussion Paper 9671, Institute of Labor Economics (IZA), Bonn, 2016.

Seema Jayachandran and Adriana Lleras-Muney. Life expectancy and human capital investments: Evidence from maternal mortality declines. *The Quarterly Journal of Economics*, 124(1):349–397, 2009. doi: 10.1162/qjec.2009.124.1.349.

David S. Lee. Training, wages, and sample selection: Estimating sharp bounds on treatment effects. *Review of Economic Studies*, 76(3):1071–1102, 2009.

David S. Lee and Thomas Lemieux. Regression discontinuity designs in economics. *Journal of Economic Literature*, 48(2):281–355, 2010. doi: 10.1257/jel.48.2.281.

Alexander Lehner. Bandwidth selection for spatial hac standard errors, 2026. URL `https://arxiv.org/abs/2603.03997`.

George Peter Murdock. *Africa: Its Peoples and Their Culture History*. McGraw-Hill, New York, 1959.

Joseph P. Romano and Michael Wolf. Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4):1237–1282, 2005.

Joseph P. Romano and Michael Wolf. Efficient computation of adjusted p-values for resampling-based stepdown multiple testing. *Statistics & Probability Letters*, 113:38–40, 2016.

Robert Shimer and Lones Smith. Assortative matching and search. *Econometrica*, 68(2):343–369, 2000. doi: 10.1111/1468-0262.00112.

Harald Tauchmann. Lee (2009) treatment-effect bounds for nonrandom sample selection. *Stata Journal*, 14(4):884–894, 2014.

Alessandra Voena. Yours, mine, and ours: Do divorce laws affect the intertemporal behavior of married couples? *American Economic Review*, 105(8):2295–2332, 2015. doi: 10.1257/aer.20120234.

Zaki Wahhaj. An economic model of early marriage. *Journal of Economic Behavior & Organization*, 152:147–176, 2018. doi: 10.1016/j.jebo.2018.06.001.

Nicholas Wilson. Child marriage bans and female schooling and labor market outcomes: Evidence from natural experiments in 17 low- and middle-income countries. *American Economic Journal: Economic Policy*, 14(3):449–477, 2022. doi: 10.1257/pol.20200008.

Appendix

This appendix contains supporting proofs, tables, and graphs.

A.1 Proofs

Proof of Proposition 1 Fix jurisdiction j and let μ, μ' be two marriage-age distributions over $\mathscr{A}$ with μ' first-order stochastically dominating μ (the shift induced by the strict regime). For $\kappa' > \kappa$, define $g(a) \equiv M(a;\kappa') - M(a;\kappa)$. By Assumption 2(ii), g is non-decreasing in a (strictly so, since $\kappa' > \kappa$), so by the definition of FOSD, $\mathbb{E}_{\mu'}[g(a)] \geq \mathbb{E}_{\mu}[g(a)]$, i.e.

$$\mathbb{E}_{\mu'}[M(a;\kappa')] - \mathbb{E}_{\mu'}[M(a;\kappa)] \;\geq\; \mathbb{E}_{\mu}[M(a;\kappa')] - \mathbb{E}_{\mu}[M(a;\kappa)].$$

The Stage 1 objective $\mathbb{E}_{a|j}[M(a;\kappa)] - c(\kappa)$ therefore has increasing differences in (κ, μ) under the FOSD order (the cost term does not depend on μ). By Assumption 2(i), $M(a;\cdot)$ is strictly concave for every a, so $\mathbb{E}_{a|j}[M(a;\kappa)]$ is strictly concave in κ (a positively weighted average of strictly concave functions); adding $-c(\kappa)$, also strictly concave since $c'' > 0$, preserves strict concavity. The Stage 1 objective is thus strictly concave in κ for every μ, so $\kappa_j^*(\mu)$ is unique. Since $M(a;\cdot)$ is continuously differentiable in κ by Assumption 2(i) and c is smooth, the objective is smooth, strictly concave, and has strictly increasing differences in (κ, μ): these are exactly the hypotheses of the strict monotone comparative statics theorem of Edlin and Shannon (1998). The unique maximizer is therefore non-decreasing in μ under the FOSD order, strictly so whenever the optimum is interior. Taking μ and μ' to be the marriage-age distributions under the lenient and strict regimes respectively gives $\kappa^*_{\text{Strict}} \geq \kappa^*_{\text{Lenient}}$, with strict inequality if the FOSD shift is strict over a set of positive probability and the optimum is interior. □

Equilibrium existence We establish existence in three steps, treating families as atomistic price-takers with respect to the jurisdiction-specific marriage-age distribution μ^j, defined over $\mathscr{A}$.

Step 1 (Stage 1) For any fixed distribution μ^j, the family's investment problem is

$$\max_{\kappa \in [0,\bar{\kappa}]} \sum_{a \in \mathscr{A}} \mu^j(a)\, M(a;\kappa) - c(\kappa).$$

By Assumption 2(i) and $c'' > 0$, the objective is continuous and strictly concave on the compact set $[0, \bar{\kappa}]$, so a maximizer exists (Weierstrass) and is unique. By Berge's Maximum Theorem the resulting selection $\kappa_j^*(\mu^j)$ is upper hemicontinuous in μ^j; uniqueness of the maximizer upgrades upper hemicontinuity to continuity in the usual way.

Step 2 (Stage 2) Given κ_j^*, the marriage-market problem is finite-horizon over $\mathscr{A}$. The acceptance decision at each age, $\mathbf{1}\{\Delta^j(a;\kappa) \geq 0\}$, is discontinuous in κ exactly where $\Delta^j(a;\kappa)$ crosses zero. To keep the backward-induction map upper hemicontinuous at these knife-edge points, we work with the relaxed acceptance correspondence

$$\alpha^j(a;\kappa) = \begin{cases} \{1\} & \Delta^j(a;\kappa) > 0, \\ [0,1] & \Delta^j(a;\kappa) = 0, \\ \{0\} & \Delta^j(a;\kappa) < 0, \end{cases}$$

the set of acceptance probabilities consistent with optimal behaviour at a tie. This is the standard convexification used to restore hemicontinuity of discontinuous best-response correspondences in matching and assignment models (e.g. Shimer and Smith (2000)). By construction α^j is upper hemicontinuous and convex-valued in κ. Two observations complete the step. First, the value function is invariant to tie-breaking: at a tie, $\Delta^j(a) = 0$ means accepting yields $U_j^f(a) = V^j(a+1) + \theta \cdot 0 = V^j(a+1)$, exactly the value of declining, so backward induction from $V^j(a_1) = \bar{V}$ delivers a unique $V^j(a)$ regardless of how indifferent women behave. Second, only the induced marriage-age distribution depends on tie-breaking. Because families are an atomistic continuum, distinct indifferent women may break ties differently, and the set of aggregate distributions $\mathscr{M}^j(\kappa) \subseteq \Delta(\mathscr{A})$ is the set of averages over measurable profiles of individually optimal (pure) acceptance choices; such aggregates over a continuum are convex by the standard Aumann-type aggregation argument. (This is the substantive content of the convexification: with finitely many agents, the map from tie-breaking probabilities at multiple ages into distributions is multilinear and its image need not be convex; aggregation over the continuum is what delivers convexity.) The correspondence $\mathscr{M}^j(\kappa)$ is therefore upper hemicontinuous and convex-valued in κ, and coincides with the single-valued map generated by $\mathbf{1}\{\Delta^j(a;\kappa) \geq 0\}$ whenever no tie occurs.

Step 3 (Stage 3) The household chooses from the finite action set $\{f_1, s_1, \ell_1\}$. Given (S_0, ω), optimal household choices and the resulting match value $M(a;\kappa_j^*)$ are well defined; this is the object whose regularity is assumed directly in Assumption 2 rather than re-derived from $\{\Pi_f, \Pi_s, \Pi_\ell\}$, since V_2 and Ω are themselves maintained rather than derived

primitives (Assumption 6).

Aggregate consistency An equilibrium requires $\mu^j \in \mathscr{M}^j(\kappa_j^*(\mu^j))$. The composite correspondence $\mu^j \mapsto \kappa_j^*(\mu^j) \mapsto \mathscr{M}^j(\kappa_j^*(\mu^j))$ maps the compact, convex simplex $\Delta(\mathscr{A})$ into itself, is convex-valued (by the convexification in Step 2) and upper hemicontinuous (the composition of a continuous function with an upper hemicontinuous, convex-valued correspondence). Kakutani's fixed-point theorem guarantees existence of μ^{j*} satisfying $\mu^{j*} \in \mathscr{M}^j(\kappa_j^*(\mu^{j*}))$.

Assumption 3, stated as a genericity condition on the model's primitives rather than on the endogenous equilibrium object, implies that for almost every parameter configuration no tie occurs at the fixed point, i.e. $\Delta^j(a;\kappa_j^*(\mu^{j*})) \neq 0$ for every $a \in \mathscr{A}$. At such a fixed point α^j is single-valued and coincides with $\mathbf{1}\{\Delta^j(a) \geq 0\}$, so μ^{j*} is a genuine, non-randomized equilibrium distribution. □

Proof of Proposition 2 and GE extension *Partial equilibrium*: Fix a and hold $V^j(a+1)$ and $M(a)$ constant. Since $\Delta^j(a) = M(a) - \Lambda^j(a) - V^j(a+1)$ is strictly decreasing in $\Lambda^j(a)$, a higher enforcement cost reduces the woman's net gain and weakly reduces the marriage hazard. Strict inequality holds when the feasibility constraint binds for one regime but not the other.

General equilibrium extension: The full GE result follows by backwards induction from $V^j(a_1) = \bar{V}$, and applies at ages where marriage occurs in at least one regime (i.e. $U_j^f(a) > V^j(a+1)$ for some j), which is the relevant case for the enforcement wedge to have bite. At $a_1 - 1$, $V^j(a_1)$ is fixed and the partial-equilibrium result applies directly. At $a_1 - 2$, the stricter wedge reduces the marriage hazard at $a_1 - 1$, which weakly lowers $V^j(a_1 - 1)$ under the strict regime (fewer marriages at $a_1 - 1$ remove the option value $U_j^f > V^j$ from the Bellman). A lower $V^j(a_1 - 1)$ enters $\Delta^j(a_1 - 2) = M(a_1 - 2) - \Lambda^j(a_1 - 2) - V^j(a_1 - 1)$ with a positive sign, partially offsetting the direct wedge effect. The net effect on $\Delta^j(a_1 - 2)$ is negative provided the direct Λ^j effect dominates this feedback, a condition satisfied whenever $\Lambda^j(a_1 - 2) \geq 0$ and the reduction in V^j is bounded by the option value removed. Restricting attention to ages below the legal threshold (where $\Lambda^j > 0$) ensures this dominance holds. The direction is preserved at each induction step, so the partial-equilibrium comparative static extends to equilibrium, resolving the circularity in the existence argument. □

Additional conditions for Assumption 6 The cross-point conditions required to close the proofs of Lemmas 1 and 2 are:

$$V_2(h_0+\phi_s, e_0+1, 0) - V_2(h_0, e_0, 1) \text{ is strictly increasing in } e_0,$$
$$V_2(h_0+\phi_\ell, e_0, 0) - V_2(h_0, e_0, 1) \text{ is strictly increasing in } h_0.$$

These conditions involve evaluations at different (h_1, e_1) points and are not implied by pointwise increasing differences alone; they are maintained assumptions.

Strict monotonicity of $\Pi_s - \Pi_f$ in e_0 (established below) pins down at most one crossing; it does not by itself guarantee that a crossing exists. Since $\kappa \in [0, \bar{\kappa}]$ and T_m is bounded on the finite age grid $\mathscr{A}$, $e_0 = e_b + \kappa T_m^\gamma$ ranges over the compact interval $[e_b,\, e_b + \bar{\kappa} T_{m,\max}^\gamma]$; h_0 is likewise bounded on its feasible support. We impose as an additional maintained condition that

$$\Pi_s(e_b) - \Pi_f(e_b) < 0 \quad \text{and} \quad \Pi_s\big(e_b + \bar{\kappa} T_{m,\max}^\gamma\big) - \Pi_f\big(e_b + \bar{\kappa} T_{m,\max}^\gamma\big) > 0,$$

and analogously for $\Pi_\ell - \Pi_f$ at the lower and upper endpoints of h_0's feasible support. Together with strict monotonicity, the intermediate value theorem then guarantees existence, and uniqueness, of the thresholds $\bar{e}$ and $\bar{h}$ on their respective compact domains. This is a condition on relative magnitudes at two fixed endpoints, not an Inada-type requirement that V_2 diverge, and so remains consistent with V_2 being bounded (Assumption 6).

Proof of Lemma 1 $\Pi_s - \Pi_f = -\omega\upsilon_f - (1-\omega)\upsilon_m - \omega\chi_s + \beta[V_2(h_0+\phi_s, e_0+1, 0) - V_2(h_0, e_0, 1)]$. The continuation term $\beta[V_2(h_0+\phi_s, e_0+1, 0) - V_2(h_0, e_0, 1)]$ is strictly increasing in e_0 by the cross-point condition above. The ω-dependent flow terms have derivative $(\upsilon_m - \upsilon_f - \chi_s) \cdot \partial\omega/\partial e_0 \geq 0$ under Assumption 5 (recall Ω is only weakly increasing in each argument), so the ω-feedback weakly reinforces the continuation term. Together, $\Pi_s - \Pi_f$ is strictly increasing in e_0; existence of the threshold $\bar{e}$ then follows from the boundary condition above via the intermediate value theorem, with uniqueness from strict monotonicity. □

Proof of Lemma 2 $\Pi_\ell - \Pi_f = \log(w^f + y_m + Rb_0) - \log(y_m + Rb_0) - \omega\chi_\ell - \omega\upsilon_f - (1-\omega)\upsilon_m + \beta[V_2(h_0+\phi_\ell, e_0, 0) - V_2(h_0, e_0, 1)]$. The wage term is strictly increasing in h_0 since $\partial w^f / \partial h_0 > 0$. The continuation term is strictly increasing in h_0 by the cross-point condition above. The ω-dependent flow terms have derivative $(\upsilon_m - \upsilon_f - \chi_\ell) \cdot \partial\omega/\partial h_0 \geq 0$

under Assumption 5, so the ω-feedback weakly reinforces the other two terms. Together, $\Pi_\ell - \Pi_f$ is strictly increasing in h_0; existence of the threshold $\bar{h}$ then follows from the boundary condition above via the intermediate value theorem, with uniqueness from strict monotonicity. □

Threshold fixed points The threshold $\bar{e}$ in Lemma 1 is implicitly ω-dependent since $\omega = \Omega(b_0, h_0, e_0; \theta)$ is weakly increasing in e_0, so the threshold condition $\Pi_s = \Pi_f$ defines a fixed-point equation in e_0. As shown in the proof of Lemma 1, the continuation term in $\Pi_s - \Pi_f$ is strictly increasing in e_0 and the ω-dependent flow term is weakly increasing: the ω-feedback contributes $(\upsilon_m - \upsilon_f - \chi_s) \cdot \partial\omega/\partial e_0 \geq 0$ under Assumption 5, weakly reinforcing rather than opposing the continuation term. The cross-point condition in Assumption 6 therefore delivers strict monotonicity of $\Pi_s - \Pi_f$ in e_0, which gives uniqueness of the crossing; existence follows from the boundary condition above together with the intermediate value theorem, since e_0 ranges over a compact interval. The same applies to $\bar{h}$ in Lemma 2. □

Proof of Lemma 3 By Lemma 1, $e_0 > \bar{e}$ implies $\Pi_s > \Pi_f$, so $f_1 = 1$ is not chosen. By Lemma 2, $h_0 > \bar{h}$ implies $\Pi_\ell > \Pi_f$, so again $f_1 = 1$ is not chosen. This establishes (i). Parts (ii) and (iii) follow by comparing Π_s and Π_ℓ directly. Monotonicity follows because $\Pi_s - \Pi_f$ is increasing in e_0 and $\Pi_\ell - \Pi_f$ is increasing in h_0. □

Proof of Lemma 4 $\partial\Pi_f/\partial\omega = \upsilon_f - \upsilon_m$ and $\partial\Pi_s/\partial\omega = -\chi_s$, so $\partial(\Pi_f - \Pi_s)/\partial\omega = \upsilon_f - \upsilon_m + \chi_s < 0$ under Assumption 5. Analogously, $\partial(\Pi_f - \Pi_\ell)/\partial\omega = \upsilon_f - \upsilon_m + \chi_\ell < 0$. Since both differences are decreasing in ω, a lower ω makes Π_f more attractive relative to the alternatives. □

Proof of Proposition 3 By Assumption 7, higher a_m implies weakly higher (b_0, h_0, e_0). Two cases arise. First, if the higher entry state satisfies $e_0 > \bar{e}$ or $h_0 > \bar{h}$, Lemmas 1–3 imply birth is not chosen; instead schooling or work is selected, accumulating higher (e_1, h_1). By Assumption 6, V_2 is increasing in (h_1, e_1), and the optimal payoff Π^* is also increasing in (h_0, e_0) through both the wage $w^f(h_0, e_0)$ and the continuation value, so M is strictly higher. Second, if both marriage ages produce entry states below both thresholds so that birth is chosen in both cases, Lemmas 1–3 do not apply. Three effects operate: higher b_0 raises consumption and hence Π_f; higher (h_0, e_0) raises $V_2(h_0, e_0, 1)$; but higher (b_0, h_0, e_0) also

raises ω, and since $\partial \Pi_f / \partial \omega = v_f - v_m < 0$, higher ω lowers Π_f. By Assumption 8, the positive b_0 and V_2 effects dominate. In both cases $M(a'; \kappa_j^*) \geq M(a; \kappa_j^*)$ for $a' \geq a$. □

Proof of Proposition 4 Part (a) follows from Assumption 7 and the monotonicity of e_0 in a_m through T_m^γ. Parts (b)–(c) follow from Lemmas 1–3: higher (e_0, h_0) implies lower Π_f relative to Π_s and Π_ℓ, shifting the period-1 choice away from birth. Part (d) follows from $\omega = \Omega(b_0, h_0, e_0; \theta)$ increasing in (b_0, h_0, e_0) combined with part (a). Part (e) follows directly: delayed marriage raises (e_1, h_1) and lowers n_1, each of which increases $\bar{V}^f$. □

Micro-foundation for Ω The monotonicity of $\Omega(b_0, h_0, e_0; \theta)$ can be derived from Nash bargaining in the household: the woman's outside option is increasing in (b_0, h_0, e_0), so her bargained share rises with her entry state. See Chiappori (1992) and Browning et al. (2014) for the collective household framework from which Ω can be micro-founded. We treat Ω as a maintained assumption here rather than deriving it from a specified period-2 bargaining problem.

Table 4 summarises the main threats to identification discussed in Section V and the design features or robustness checks that address each one; it appears in Appendix A.2 alongside the data documentation it references.

A.2 Data Appendix: Pixel-Level and Auxiliary Geographic Data

A2.0 DHS Cluster Locations, Borders, and Spatial Merging *Survey Cluster Coordinates:* All individual-level data come from the Demographic and Health Surveys (DHS). Respondents are sampled in survey clusters (primary sampling units), and every woman in a cluster shares its single GPS coordinate. To protect confidentiality, the DHS randomly displaces published cluster coordinates by 0–2 km in urban areas and 0–5 km in rural areas, with one percent of rural clusters displaced up to 10 km. Clusters whose GPS collection failed are published at coordinates $(0, 0)$ and are excluded by the bandwidth restriction. All spatial variables below are measured at the published (displacement-protected) cluster coordinate.

Treatment Assignment: Each respondent's side of the border is assigned from the country of the DHS survey in which she was interviewed, not from the geometric position of her displaced coordinate. This prevents GPS displacement from misclassifying respondents near the border onto the wrong side.

National Borders: Country boundaries are taken from GADM version 4 (level 0). For each of the twelve border pairs, the study border is the shared boundary of the two countries' polygons, computed in a pair-specific Albers equal-area projection.

Pixel Grid: Following Michalopoulos and Papaioannou (2013, 2014), each country pair is overlaid with a 12.5 km $\times$ 12.5 km grid used to assign the PRIO-GRID-based controls (soil suitability, malaria index, distance to capital, water area, diamonds, petroleum) to survey clusters.

Assignment to a Unique Border: A survey cluster can lie within the estimation bandwidth of more than one study border (for example in the Namibia–Zambia–Zimbabwe border region). Each cluster is assigned only to its nearest study border, so no respondent appears in more than one border-pair comparison.

Cohort Restriction: The estimation sample is restricted to women whose exposure to the legal discontinuity was complete. For each border pair, the at-risk window runs from the control country's minimum legal marriage age with parental consent to the treated country's, this being the range of ages over which the two countries' laws differ: 15–18 in most pairs, 16–18 or 17–18 where the control age is higher, and 15–21 for Rwanda–Tanzania. A respondent is retained only if her entire at-risk window fell within the years the legal gap was in force, so that she was subject to the differential regime throughout the period in which she could have married as a minor. Women whose window only partially overlapped the gap passed some of those ages under identical law on both sides of the border, and retaining them would dilute the treatment. Of women within the 200 km export bandwidth, 85.7 percent satisfy this condition, 10.6 percent are only partially exposed, and 3.7 percent are not exposed at all.

A2.1 Pixel-Level Data Using the GPS coordinates of DHS survey clusters, we augment the individual-level data with detailed geographic information. Controls 1–7 are extracted directly at each respondent's (displacement-protected) cluster coordinate from global rasters (light density at 30 arc-second resolution; controls 2–7 at 0.5° resolution); controls 8–11 are assigned through a 12.5 km $\times$ 12.5 km grid following Michalopoulos and Papaioannou (2013, 2014).

Light Density: Nighttime light intensity in 2022, from the harmonized DMSP–OLS/VIIRS satellite composite series of Li, Zhou, Zhao and Zhao (2020), extracted at the cluster coordinate from the native 30 arc-second ($\approx$1 km) raster.

Population Density: Population per square kilometre in 2020, computed as gridded population counts divided by true cell area. Source: Global Human Settlement Layer (GHSL), European Commission Joint Research Centre.

Elevation: Average elevation in metres. Source: TerrainBase global digital elevation model, NOAA and U.S. National Geophysical Data Center, via the PRIO-GRID raster collection (Tollefsen, Strand and Buhaug 2012).

Travel Time to Market: Average travel time in minutes to the nearest significant settlement, from the global accessibility surface distributed with the PRIO-GRID raster collection.

Rainfall: Long-run mean annual precipitation, averaged over all available years of the Climatic Research Unit time series (CRU TS), University of East Anglia.

Temperature: Long-run mean annual temperature in degrees Celsius, averaged over 1982–2023 from the CRU TS temperature series.

Distance to Coast: Straight-line distance in kilometres from the cluster coordinate to the nearest ocean cell, computed from the Natural Earth land/sea mask.

Soil Suitability: Average land quality for cultivation (0–1 scale), from Ramankutty, Foley, Norman and McSweeney's global agricultural suitability dataset (SUIT), distributed in ESRI GRID format by the Center for Sustainability and the Global Environment (SAGE), University of Wisconsin.

Malaria Index: Malaria stability index measuring climatic and vector suitability for transmission, from Kiszewski, Mellinger, Spielman, Malaney, Sachs and Sachs (2004).

Distance to Capital: Distance in kilometres from the pixel centroid to the national capital city of the corresponding country. Constructed using PRIO-GRID (Tollefsen, Strand and Buhaug 2012), with capital-city coordinates from the cShapes dataset (Weidmann, Kuse and Gleditsch 2010).

Water Area: Percent of the pixel covered by inland water bodies and wetlands. Source: Global Lakes and Wetlands Database (GLWD v2.0).

Distance to Border (running variable): Distance in kilometres from the cluster coordinate to the nearest point on the relevant international border, computed from GADM version 4 national boundaries in a pair-specific equal-area projection; signed positive on the treated (stricter female marriage-law) side.

A2.2 Auxiliary Geographic Data (Balance Tests and Robustness) *Diamond and Petroleum Deposits:* Indicator variables equal to one if a diamond mine (resp. on-shore oil or gas field) lies in the pixel. Sources: PRIO Diamond Resources dataset; Petroleum Dataset v1.2 (Lujala, Rod and Thieme 2007). Petroleum takes the value zero everywhere within 100 km of the twelve study borders; diamond deposits appear in 0.2 percent of the sample.

Distance to Protestant and Catholic Missions: Distance in kilometres to the nearest historical Christian mission station, digitised from Roome's (1924) missionary atlas (933 Protestant and British and Foreign Bible Society stations; 361 Catholic stations), as used by Nunn and Wantchekon (2011).

Distance to Colonial Railways and Explorer Trade Routes: Distance in kilometres to the nearest colonial-era railway line and to the nearest pre-colonial explorer route, from the digitised maps used in Nunn and Wantchekon (2011) and Ciolek (2001).

Country-Level Geography: GDP per capita (2000, World Bank), terrain ruggedness, distance to coast, and geographic coordinates at the country level, from the Nunn and Puga (2012) replication data. These variables are reported in the covariate balance table but are not included as regression controls: the treatment is itself a country-level assignment within each border pair, and with twelve pairs the design does not support separating the marriage-law discontinuity from other country-level attributes.

Ethnic Homelands: Ancestral ethnic homeland boundaries from the digitised Murdock (1959) map of African peoples (844 polygons), as constructed for Nunn and Wantchekon (2011). Homeland identity defines the ethnicity fixed effects; 32 homelands are split by one of the twelve study borders. Bride-price customs (Ethnographic Atlas variable EA006, via D-PLACE) are assigned to homelands by spatial match, with nearest-neighbour imputation for uncoded groups.

Table 4: Identification Threats and Responses

Threat	Why it matters	How it is addressed
Ethnic and cultural differences	Marriage norms, preferences, and institutions may differ systematically across the border	Murdock (1959) ethnicity fixed effects compare women within the same pre-colonial homeland on opposite sides of the border; results are robust to replacing these with GREG ethnic fixed effects (Table 19)
Development gradients	Richer or better-served areas may have better outcomes independently of legal regimes	Eleven pixel-level controls, including nighttime lights, population density, agricultural suitability, malaria ecology, and distance to capital (Appendix A.2); estimates robust to dropping all of them (Table 13)
Smooth spatial trends	Outcomes may vary continuously with distance from the border, confounding the discontinuity	Border-pair-specific linear slopes and kinks in distance from the cutoff; robust to 2nd- and 3rd-order polynomials as well (Tables 14 and 15)
Policy bundle confound	Other child-protection laws (compulsory schooling, minimum employment age) may co-move with marriage age laws	WORLD Policy Analysis Center data show no systematic co-movement across all 12 border pairs; Namibia pairs with co-moving compulsory-schooling laws excluded as robustness check (Table 12) — this materially reduces the education effect, so compulsory schooling is treated as a contributing rather than a fully ruled-out channel; Niger's single border pair is addressed via the leave-one-border-pair-out check (Table 16)
Migration around marriage	Women may cross the border to marry under a more favourable legal regime, violating the residence-based identification assumption	Migration robustness excludes women who migrated within three years before their marriage (Table 9)
Spatially correlated errors	Residuals may be correlated with nearby observations, understating standard errors	Primary clustering at the DHS survey-cluster level (≈700 clusters); Conley spatial standard errors as robustness check (Table 11)
Multiple testing	Significance across many outcomes may reflect false discoveries rather than true treatment effects	Romano-Wolf stepdown correction controls the family-wise error rate within three pre-specified outcome families; results are robust (Table 10)

Notes: Each row states a potential threat to the identifying assumption that the border discontinuity in minimum legal marriage age is the only discontinuous determinant of outcomes at the cutoff. The right column describes the design feature or robustness check that addresses the threat.

A.3 Sample description and diagnostics

Covariate Balance Table 5 reports balance tests for predetermined covariates at the RD cutoff. Each row reports the coefficient on treatment from a separate regression of the covariate on treatment, using the same border-pair fixed effects, border-pair-specific RD slopes, and treated-side kinks as the main specification, estimated on observations within 100 km of the border. A substantial number of covariates show statistically significant differences between the treated and control sides. Development- and market-access-related

covariates tend to favor the treated side: population density, luminosity, and distance to the nearest trade route are all significantly higher, distance to the nearest Protestant and Catholic mission are higher at the 10 percent level, and travel time to the nearest market and distance to the nearest colonial railway are significantly shorter, consistent with the development gradient discussed elsewhere in the paper. Environmental covariates also tend to favor the treated side: water area, the malaria ecology index, rainfall, and temperature are all significantly lower, while distance to the coast is significantly greater at the 10 percent level. Wife's age is marginally lower on the treated side, significant at the 10 percent level. Elevation, distance to the capital, and the presence of diamond deposits show no statistically significant imbalance. Three country-level covariates are also imbalanced: GDP per capita and terrain ruggedness are significantly higher on the treated side, and the country-level distance to coast is significantly lower (distinct from the pixel-level distance-to-coast row, which is measured at each respondent's own coordinate and is significantly greater on the treated side). Because these three variables take only two values within each border pair (one per country), the estimated coefficients are within-pair country differences whose clustered standard errors are likely understated, so these rows are reported as descriptive rather than as formal balance tests.

Two points bear on the interpretation of these imbalances. First, and most directly, we re-ran all main specifications after dropping the pixel-level geographic controls and ethnicity fixed effects entirely. The estimates are broadly similar to those in Table 1, suggesting that the imbalances documented in Table 5 are not mechanically driving the results. These robustness results are reported in Table 13. Second, because wife's age is only marginally imbalanced at the cutoff, cohort composition across the border does not appear to be a first-order concern; the individual-level age control in X_{ibet} is nonetheless retained throughout as a precaution. Given how many covariates show some imbalance, we treat Table 5 as motivating the breadth of the control vector used throughout, rather than as evidence that any single covariate alone explains the reduced-form results.

Table 5: Estimations of Control Variables (RD Balance Test)

Variable	Coefficient on Treated
Age	-0.547^{**}
	(0.265)
Population density	32.150^{***}
	(6.271)
Luminosity	4.536^{***}
	(1.282)
Water area	-5.416^{***}
	(1.629)
Elevation	0.131
	(6.846)
Soil suitability	-0.041^{***}
	(0.010)
Malaria index	-1.843^{***}
	(0.184)
Distance to capital	-1.787
	(15.626)
Travel time to market	-22.536^{**}
	(10.957)
Rainfall	-28.994^{***}
	(7.831)
Temperature	-0.137^{***}
	(0.041)
Distance to coast	11.777^{*}
	(6.900)
Diamond deposit in pixel	0.000
	(0.000)
Distance to colonial railway	-13.196^{***}
	(4.777)
Distance to Protestant mission	7.872
	(4.995)
Distance to Catholic mission	11.480^{*}
	(6.803)
Distance to trade route	9.191^{***}
	(3.031)
GDP per capita (country-level)	1530.730^{***}
	(83.909)
Terrain ruggedness (country-level)	0.342^{***}
	(0.038)
Distance to coast (country-level)	-0.202^{***}
	(0.022)

Notes: Each row reports the coefficient (standard error) on treatment from a separate regression of the covariate on treatment, including border-pair fixed effects, border-pair-specific RD slopes, and treated-side kinks in distance to the border. Standard errors clustered at the DHS-cluster level (country $\times$ survey-round $\times$ PSU) in parentheses. Estimates use observations within 100 km of the border cutoff. GDP per capita, terrain ruggedness, and the country-level distance to coast are country-level variables (distinct from the pixel-level Distance to coast row above, which is measured at each respondent's own coordinate): within a border pair they take only two values, so these rows are within-pair country differences whose clustered standard errors are understated; they should be read as descriptive rather than as formal inference. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Sample and legal environment Table 6 lists all twelve border pairs. For each pair, the treatment country has the higher minimum legal marriage age. The exposure period is defined by the availability of DHS survey waves that overlap with the period during which the legal age difference was in place. For RWA–TZA, Rwanda raised its minimum age to 21 in 2016; identification for this pair uses only cohorts aged 15–18 during the pre-reform period, so the legal contrast at the 18-year margin is consistent across all pairs.

Table 6: Border Pairs, Legal Marriage Age Differences, and Exposure Period

Border pair	Treatment country	Control country	Period	Treatment age	Control age
AGO–NAM	Namibia	Angola	1996–2023	18	15
BEN–BFA	Benin	Burkina Faso	2004–2023	18	17
BEN–NER	Benin	Niger	2004–2023	18	15
BFA–CIV	Côte d'Ivoire	Burkina Faso	1995–2023	18	17
BFA–GHA	Ghana	Burkina Faso	1999–2023	18	17
CIV–MLI	Côte d'Ivoire	Mali	1995–2023	18	15
BDI–TZA	Burundi	Tanzania	1995–2023	18	15
NAM–ZAF	Namibia	South Africa	1996–2023	18	15
NAM–ZMB	Namibia	Zambia	1996–2021	18	16
NAM–ZWE	Namibia	Zimbabwe	1996–2021	18	16
RWA–TZA	Rwanda	Tanzania	1995–2023	21	15
TZA–UGA	Uganda	Tanzania	1995–2023	18	15

Notes: Treatment country has the higher minimum legal marriage age (Treatment age). Control country has the lower minimum legal marriage age (Control age). For RWA–TZA, Rwanda raised its minimum age to 21 in 2016; identification uses only cohorts aged 15–18 in the pre-reform period.

Descriptive statistics Tables 7 and 8 report mean and median summary statistics by country. Treatment countries (minimum age ≥ 18) have lower rates of child marriage, later age at first birth, and higher education on average than control countries, consistent with the direction of the main results. The variation across countries in observation counts reflects differences in the number of DHS survey waves and the geographic extent of the sample within each bandwidth. We note that Niger is the smallest country sample (4,513 observations) and has the highest pre-18 marriage rate in the sample (73%). The results are robust to excluding the Benin–Niger pair, as shown in the leave-one-border-pair-out check (Table 16).

Table 7: Mean Summary Statistics by Country (Full Country Samples)

Country	Obs.	Age	Married	Before 18	First birth	Educ. (yrs)	Employed
Angola	22,110	27.30	0.71	0.32	18.57	5.69	0.57
Benin	14,261	23.44	0.74	0.35	18.85	3.62	0.67
Burkina Faso	30,511	26.16	0.85	0.47	18.87	2.59	0.69
Burundi	17,836	26.62	0.70	0.20	20.40	4.50	0.77
Cote d'Ivoire	19,593	27.81	0.75	0.34	19.03	3.36	0.63
Ghana	19,791	26.84	0.67	0.24	20.24	7.29	0.73
Mali	42,705	26.33	0.90	0.57	18.25	2.33	0.48
Namibia	11,596	24.02	0.34	0.10	19.36	8.84	0.38
Niger	4,513	21.61	0.90	0.73	17.26	1.91	0.18
Rwanda	33,319	26.78	0.61	0.10	21.51	5.30	0.73
South Africa	5,643	27.62	0.37	0.05	20.29	10.68	0.31
Tanzania	28,412	27.43	0.79	0.33	19.31	6.77	0.69
Uganda	25,457	25.41	0.81	0.43	18.27	6.30	0.75
Zambia	35,686	27.34	0.75	0.35	18.59	7.37	0.50
Zimbabwe	17,816	24.55	0.75	0.31	19.15	9.32	0.38
Control avg.		26.53	0.79	0.41	18.74	5.31	0.54
Treatment avg.		26.10	0.68	0.25	19.83	5.51	0.69

Notes: Mean values by country, full national DHS samples (all women 15–49 in the survey rounds used, not restricted to the border-bandwidth RD estimation sample). Treatment countries have minimum legal marriage age ≥ 18; control countries have minimum legal marriage age < 18. Child marriage (Before 18) is defined as first marriage or cohabitation before age 18 and is defined for all women (never-married adult women are coded 0). Employed is one minus the not-working indicator. Age is respondent age at interview (15–49).

Table 8: Median Summary Statistics by Country (Full Country Samples)

Country	Obs.	Age	Married	Before 18	First birth	Educ. (yrs)	Employed
Angola	22,110	26	1	0	18	6	1
Benin	14,261	23	1	0	19	0	1
Burkina Faso	30,511	25	1	0	18	0	1
Burundi	17,836	26	1	0	20	4	1
Cote d'Ivoire	19,593	27	1	0	18	0	1
Ghana	19,791	26	1	0	20	9	1
Mali	42,705	25	1	1	18	0	0
Namibia	11,596	23	0	0	19	9	0
Niger	4,513	22	1	1	17	0	0
Rwanda	33,319	26	1	0	21	5	1
South Africa	5,643	27	0	0	20	11	0
Tanzania	28,412	26	1	0	19	7	1
Uganda	25,457	24	1	0	18	6	1
Zambia	35,686	26	1	0	18	7	0
Zimbabwe	17,816	24	1	0	19	10	0

Notes: Median values by country, full national DHS samples (all women 15–49 in the survey rounds used, not restricted to the border-bandwidth RD estimation sample). Age refers to respondent age at interview (15–49). Child marriage (Before 18) is defined as marriage before age 18 and is defined for all women (never-married adult women are coded 0). Medians are computed using available observations for each variable.

A.4 Robustness checks

Tables 9–13 present five of the robustness checks summarised in Section VII: excluding recent migrants, Romano-Wolf multiple-hypothesis correction, Conley spatial standard errors under a data-driven bandwidth, excluding the Namibia border pairs, and excluding the geographic and ethnicity controls. The migration, Romano-Wolf, Conley, and no-controls checks preserve the sign and significance pattern of the main reduced-form estimates in Table 1 throughout. The Namibia-exclusion check is the one exception among these five, discussed below, where the continuous education-years coefficient loses significance at the 50 km bandwidth. Table 11 uses R, which is why the point estimates are marginally different from Table 1, as all other results are done via Stata.

Controlling for migration A potential concern is that individuals migrate across borders specifically to contract an underage marriage, which would then contaminate our geographic discontinuity. To address this, Table 9 excludes individuals from the sample who migrated within three years before their marriage.

The pattern is nearly unchanged. Our general pattern in education, fertility and labour market outcomes are stable. The sample drops from approximately 69,502 to 62,854 observations at the 100 km bandwidth (using the education-years estimation sample as a benchmark), indicating that a meaningful number of migrants were excluded while the main pattern remains similar.

Table 9: Reduced-Form Effects, Restricting to Non-Migrants

	Bandwidth (km)		
Outcome	50	100	200
Human capital			
Education (years)	1.000***	1.219***	1.310***
	(0.230)	(0.199)	(0.182)
Education level	0.289***	0.336***	0.335***
	(0.041)	(0.036)	(0.033)
Fertility			
Age at first birth	0.863***	0.916***	0.735***
	(0.155)	(0.123)	(0.105)
Children ever born	−0.264***	−0.327***	−0.292***
	(0.073)	(0.057)	(0.048)
Age at first sex	0.537***	0.634***	0.490***
	(0.128)	(0.110)	(0.092)
Household and labour market			
Multiple unions	0.015	−0.006	−0.009
	(0.012)	(0.009)	(0.008)
Not working	0.102***	0.117***	0.155***
	(0.028)	(0.022)	(0.019)
Domestic work	0.009*	−0.014***	−0.005
	(0.005)	(0.005)	(0.004)
White-collar (extensive)	0.024***	0.029***	0.027***
	(0.009)	(0.008)	(0.007)
White-collar (intensive)	0.030**	0.043***	0.044***
	(0.012)	(0.011)	(0.010)

Notes: The sample excludes women who migrated within three years before their marriage. Standard errors clustered at the DHS survey-cluster level in parentheses. All other specifications identical to the main reduced-form table. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Romano-Wolf multiple-hypothesis correction As the results span multiple outcome families, a natural concern is that some significant findings may arise from testing mul-

tiple outcomes rather than genuine treatment effects. To address this, Table 10 reports Romano-Wolf stepdown p-values (Romano and Wolf, 2005, 2016) for the reduced-form estimates at the 100 km bandwidth, adjusted within three pre-specified outcome families: education, fertility, and labour-market outcomes. The Romano-Wolf procedure controls the family-wise error rate under arbitrary dependence across outcomes within each family, making it more conservative than uncorrected p-values.

The adjustment leaves the core findings intact. Within the education family, both outcomes retain significance after correction. Within fertility, age at first birth, children ever born, and age at first sex remain significant, while multiple unions is insignificant both before and after correction, consistent with their pattern in the main reduced-form results (Table 1). In the labour-market family, not working, domestic work, and both white-collar outcomes all remain significant after correction. The close correspondence between the uncorrected model p-values and the Romano-Wolf p-values across all three families reduces concern that the findings are driven by multiple testing.

Table 10: Romano-Wolf Multiple-Hypothesis Correction (100 km Bandwidth)

Outcome	Model p	Resample p	Romano-Wolf p
Panel A: Education outcomes			
Education (years)	0.000	0.001	0.001^{***}
Education level	0.000	0.001	0.001^{***}
Panel B: Fertility outcomes			
Age at first birth	0.000	0.001	0.001^{***}
Children ever born	0.000	0.001	0.001^{***}
Age at first sex	0.000	0.001	0.001^{***}
Multiple unions	0.310	0.188	0.188
Panel C: Labour-market outcomes			
Not working	0.000	0.001	0.001^{***}
Domestic work	0.001	0.001	0.001^{***}
White-collar (extensive)	0.000	0.001	0.001^{***}
White-collar (intensive)	0.000	0.001	0.001^{***}

Notes: Romano-Wolf stepdown-adjusted p-values for the reduced-form coefficient on treated at the 100 km bandwidth, controlling the family-wise error rate within three pre-specified outcome families (Panels A–C) under arbitrary within-family dependence. All regressions include border-pair fixed effects, border-pair-specific slopes and kinks, geographic controls, ethnicity fixed effects, survey-year fixed effects, and individual controls. Model p is the uncorrected p-value from clustered standard errors; Resample p is the individual bootstrap p-value prior to the stepdown adjustment; Romano-Wolf p is the final family-wise-error-rate-controlled p-value. Computed via cluster bootstrap resampling of DHS clusters (1,000 replications). Stars are based on the Romano-Wolf p-value: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Conley spatial standard errors Women near each other on opposite sides of the border may have correlated residuals even after controlling for ethnicity and geography, as they may share the same local labour market, infrastructure, and unobserved neighbourhood characteristics. Standard clustering at the DHS survey-cluster level accounts for within-cluster correlation but does not allow for cross-cluster spatial dependence.

We address this using a data-driven cutoff selected from the empirical residual covariogram (Lehner, 2026), specifically the first distance bin at which the smoothed residual covariance becomes non-positive, and reported in Table 11. This lets the data determine the appropriate radius rather than imposing one. Lehner (2026) shows that the relationship between the bandwidth and the magnitude of spatial HAC standard errors is inverse-U shaped, implying that both too narrow and too wide bandwidths can lead to underestimated standard errors. This motivates using a data-driven selector rather than defaulting to an

arbitrarily chosen fixed radius.

Education level, age at first birth, children ever born, age at first sex, not working, and both white-collar measures retain significance at the 1 percent level under the data-driven spatial correction at the main bandwidth. Domestic work, which is not significant with DHS-clustered standard errors alone, also becomes significant at the 5 percent level at the 100 km bandwidth under this correction. Education in years retains significance at the 1 percent level across all three bandwidths; its standard error (0.408 at the 100 km bandwidth) is roughly double the DHS-clustered standard error (0.193), but the coefficient remains comfortably significant.

Table 11: Reduced-Form Estimates with Data-Driven Conley Spatial Standard Errors

	Bandwidth (km)		
Outcome	50	100	200
Human capital			
Education (years)	0.955***	1.239***	1.281***
	(0.317)	(0.412)	(0.355)
Education level	0.281***	0.341***	0.334***
	(0.055)	(0.057)	(0.068)
Fertility			
Age at first birth	0.864***	0.948***	0.752***
	(0.153)	(0.128)	(0.124)
Children ever born	−0.260***	−0.326***	−0.261***
	(0.082)	(0.074)	(0.074)
Age at first sex	0.528***	0.651***	0.475***
	(0.159)	(0.134)	(0.155)
Household and labour market			
Multiple unions	0.009	−0.012	−0.011
	(0.011)	(0.009)	(0.009)
Not working	0.103***	0.113***	0.146***
	(0.034)	(0.030)	(0.033)
Domestic work	0.006	−0.015**	−0.008
	(0.006)	(0.006)	(0.007)
White-collar (ext.)	0.026***	0.031***	0.027***
	(0.009)	(0.011)	(0.008)
White-collar (int.)	0.034***	0.046***	0.045***
	(0.012)	(0.014)	(0.012)

Notes: Reduced-form estimates. The coefficient is the estimated effect of residence on the strict-law side of the border. Standard errors in parentheses are Conley spatial-HAC standard errors using a data-driven spatial cutoff selected from the empirical residual covariogram (Lehner, 2026). The selected cutoff is the first distance bin at which the smoothed residual covariance becomes non-positive, with residuals aggregated to the DHS-cluster level. All regressions include border-pair fixed effects, border-pair-specific slopes and kinks, pixel-level geographic controls, ethnicity fixed effects, and survey-year fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Excluding Namibia border pairs Four border pairs (Angola–Namibia, Namibia–South Africa, Namibia–Zambia, and Namibia–Zimbabwe) share Namibia as the treatment country that has both a stricter minimum marriage age and a stricter compulsory schooling age than its control counterparts. This raises the concern that better outcomes near Namibia reflect the stronger education environment rather than the higher marriage age. If compulsory schooling were the driver, dropping these pairs should attenuate the education estimates.

Table 12 shows exactly this pattern: the education-years effect without Namibia falls to 0.63 years at the 100 km bandwidth, roughly half the pooled estimate of 1.21 years in Table 1, and is no longer statistically significant at the 50 km bandwidth. This indicates that the Namibia border pairs, where compulsory schooling is also stricter on the treatment side, account for a meaningful share of the aggregate education effect, so the education result cannot be fully attributed to the marriage-age discontinuity alone. The education effect outside the Namibia pairs remains positive at all three bandwidths and significant at the 100 and 200 km bandwidths, so the mechanism is not solely a Namibia artifact, but compulsory schooling should be read as a contributing rather than a ruled-out channel. Fertility and white-collar results are comparatively stable in sign and significance across bandwidths.

Niger also presents a concern. As a control country, Niger has weaker laws across multiple dimensions (no formal property rights for women, no right to own a bank account, and a low age of consent), meaning the control side in the Benin–Niger pair is doubly disadvantaged relative to the treatment side. This could exaggerate the apparent treatment effect by attributing to the marriage-age law what is in fact a broader legal disadvantage on the control side. Because Niger contributes only a single border pair, we do not report a separate exclusion table for it; instead, the leave-one-border-pair-out check (Table 16) directly addresses this concern by dropping Benin–Niger alone. The sign and significance pattern is preserved across all outcomes, with point estimates close to the pooled estimate, suggesting the results are not driven by Niger's outlier legal environment.

Using data from the World Bank's Women, Business and the Law indicators and the WORLD Policy Analysis Center, we assess whether treatment and control countries differ systematically in their broader legal environment for women across five dimensions: age of consent, minimum compulsory education, women's property rights, the right to work outside the home, and the right to own a bank account. Table 21 summarises the comparisons across all 12 border pairs. The pattern is broadly balanced and does not show a systematic advantage on the treatment side. These issues are highlighted in Wilson (2022), which we make sure to account for.

Table 12: Reduced-Form Effects, Excluding Namibia Border Pairs

	Bandwidth (km)		
Outcome	50	100	200
Human capital			
Education (years)	0.217	0.687***	0.620***
	(0.233)	(0.207)	(0.188)
Education level	0.180***	0.268***	0.240***
	(0.044)	(0.039)	(0.035)
Fertility			
Age at first birth	0.677***	0.899***	0.714***
	(0.183)	(0.144)	(0.125)
Children ever born	−0.211***	−0.282***	−0.181***
	(0.080)	(0.064)	(0.053)
Age at first sex	0.420***	0.559***	0.324***
	(0.142)	(0.124)	(0.105)
Household and labour market			
Multiple unions	0.016	−0.005	−0.014*
	(0.013)	(0.010)	(0.008)
Not working	0.042	0.048**	0.083***
	(0.029)	(0.023)	(0.021)
Domestic work	0.001	−0.020***	−0.011**
	(0.005)	(0.005)	(0.004)
White-collar (extensive)	0.021**	0.032***	0.023***
	(0.009)	(0.008)	(0.007)
White-collar (intensive)	0.024**	0.041***	0.032***
	(0.012)	(0.010)	(0.010)

Notes: The four Namibia border pairs are excluded from the sample. Standard errors clustered at the DHS survey-cluster level in parentheses. All other specifications identical to the main reduced-form table. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Excluding geographic controls The main specification includes eleven pixel-level geographic controls (Appendix A.2), and Table 5 documents imbalances across these along with a further set of historical and country-level covariates (diamond deposits, distance to colonial railways and historical missions, distance to pre-colonial trade routes, and country-level GDP and ruggedness) that are reported for balance purposes but not included in the regression controls. Because several covariates in the control vector itself show significant imbalance, one might worry that the results are sensitive to the inclusion of the geographic control vector generally, and may be driving our results. One could make a similar ar-

gument for our ethnicity fixed effects. We run our results without the eleven pixel-level controls and without ethnicity fixed effects.

Point estimates are somewhat smaller without geographic controls, particularly for education and fertility outcomes, suggesting that the excluded covariates absorb some variation correlated with both treatment and the outcomes in the same direction as the treatment effect. The sign and significance pattern is preserved across nearly all outcomes and bandwidths, with the exception of multiple unions, which becomes significant at the 5 percent level at the 200 km bandwidth without the control vector. Overall, this confirms that the main results are not mechanically driven by the geographic and ethnicity controls.

Table 13: Reduced-Form Effects, Excluding Geographic Controls and Ethnicity Fixed Effects

	Bandwidth (km)		
Outcome	50	100	200
Human capital			
Education (years)	0.984***	1.213***	1.014***
	(0.231)	(0.179)	(0.141)
Education level	0.296***	0.346***	0.304***
	(0.042)	(0.033)	(0.026)
Fertility			
Age at first birth	0.782***	0.780***	0.762***
	(0.150)	(0.104)	(0.080)
Children ever born	−0.282***	−0.307***	−0.284***
	(0.069)	(0.052)	(0.040)
Age at first sex	0.398***	0.463***	0.416***
	(0.120)	(0.090)	(0.070)
Household and labour market			
Multiple unions	0.009	−0.007	−0.012**
	(0.011)	(0.008)	(0.006)
Not working	0.096***	0.109***	0.108***
	(0.026)	(0.018)	(0.015)
Domestic work	0.004	−0.001	0.007**
	(0.005)	(0.004)	(0.003)
White-collar (extensive)	0.031***	0.032***	0.026***
	(0.009)	(0.006)	(0.005)
White-collar (intensive)	0.045***	0.049***	0.041***
	(0.012)	(0.009)	(0.007)

Notes: Pixel-level geographic controls (nighttime light intensity, population density, agricultural suitability, malaria ecology, distance to capital, water/wetland percent, elevation, travel time to market, rainfall, temperature, and distance to coast) are excluded from the control vector. Ethnicity fixed effects are also excluded. Standard errors clustered at the DHS survey-cluster level in parentheses. All other specifications identical to the main reduced-form table. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Alternative RD polynomial orders The main specification fits a border-pair-specific linear polynomial with a kink at the cutoff. Because the choice of polynomial order in an RD design is itself a specification decision, we re-estimate the main results replacing the linear-with-kink polynomial with a 2nd-order and a 3rd-order border-pair-specific polynomial in distance to the border, each interacted with treatment. Tables 14 and 15 report the results.

At the 100 km bandwidth, the bandwidth used for the main specification, every outcome

is stable in sign and significance across polynomial orders. Education (years), education level, age at first birth, children ever born, not working, domestic work, and both white-collar outcomes remain significant at the 1 percent level at every order, with point estimates close to the linear-specification values in Table 1. Age at first sex is the one outcome more sensitive to polynomial order at this bandwidth, weakening from 0.650^{***} to 0.637^{***} and then 0.363^{***}, though it remains significant at the 5 percent level throughout. Multiple unions remain statistically indistinguishable from zero at every order, consistent with the main results.

At the 50 km and 200 km bandwidths the pattern is somewhat less clean: under the 3rd-order polynomial, children ever born loses significance at 50 km, domestic work loses significance at 200 km, and white-collar (extensive) weakens to the 5 percent level at 50 km. These losses are concentrated in coefficients that were already close to conventional significance thresholds in the linear specification, and none of them affect the 100 km results that anchor the paper's main estimates. This pattern is typical of higher-order polynomial RD specifications, which trade a more flexible fit near the cutoff for wider standard errors; it does not point to a specific misspecification of the baseline linear-with-kink polynomial, but age at first sex, and the 50/200 km estimates more broadly, should be read as somewhat less robust to functional-form choice than the 100 km headline results.

Table 14: Reduced-Form Effects, 2nd-Order RD Polynomial

	Bandwidth (km)		
Outcome	50	100	200
Human capital			
Education (years)	1.013***	1.140***	1.339***
	(0.280)	(0.226)	(0.200)
Education level	0.299***	0.324***	0.358***
	(0.052)	(0.041)	(0.036)
Fertility			
Age at first birth	0.927***	0.954***	0.826***
	(0.199)	(0.147)	(0.119)
Children ever born	−0.208**	−0.354***	−0.279***
	(0.087)	(0.068)	(0.056)
Age at first sex	0.382**	0.609***	0.595***
	(0.162)	(0.125)	(0.104)
Household and labour market			
Multiple unions	0.024*	−0.002	−0.003
	(0.014)	(0.011)	(0.008)
Not working	0.093***	0.116***	0.116***
	(0.035)	(0.026)	(0.022)
Domestic work	0.003	−0.014**	−0.012***
	(0.005)	(0.006)	(0.004)
White-collar (extensive)	0.029***	0.027***	0.033***
	(0.011)	(0.008)	(0.007)
White-collar (intensive)	0.040***	0.043***	0.050***
	(0.015)	(0.012)	(0.010)

Notes: The border-pair-specific RD polynomial in distance to the border is replaced with a 2nd-order polynomial interacted with treatment (the main specification uses a 1st-order polynomial with a kink). Standard errors clustered at the DHS survey-cluster level in parentheses. All other specifications identical to the main reduced-form table. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 15: Reduced-Form Effects, 3rd-Order RD Polynomial

	Bandwidth (km)		
Outcome	50	100	200
Human capital			
Education (years)	1.094***	0.950***	1.108***
	(0.343)	(0.262)	(0.219)
Education level	0.311***	0.272***	0.302***
	(0.062)	(0.049)	(0.040)
Fertility			
Age at first birth	0.861***	0.797***	0.736***
	(0.253)	(0.182)	(0.137)
Children ever born	−0.101	−0.252***	−0.257***
	(0.107)	(0.079)	(0.065)
Age at first sex	0.391*	0.363**	0.514***
	(0.208)	(0.152)	(0.117)
Household and labour market			
Multiple unions	0.034**	0.019	−0.009
	(0.017)	(0.013)	(0.010)
Not working	0.125***	0.123***	0.115***
	(0.043)	(0.032)	(0.025)
Domestic work	0.006	−0.016**	−0.008
	(0.008)	(0.006)	(0.005)
White-collar (extensive)	0.025*	0.027***	0.032***
	(0.013)	(0.010)	(0.008)
White-collar (intensive)	0.037**	0.046***	0.052***
	(0.019)	(0.014)	(0.011)

Notes: The border-pair-specific RD polynomial in distance to the border is replaced with a 3rd-order polynomial interacted with treatment (the main specification uses a 1st-order polynomial with a kink). Standard errors clustered at the DHS survey-cluster level in parentheses. All other specifications identical to the main reduced-form table. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Leave-one-border-pair-out A final concern is that the pooled reduced-form estimates in Table 1 could be driven disproportionately by a single border pair, particularly given that the twelve pairs vary in sample size, legal-age gap, and country-level conditions. To check this, we re-estimate the main specification twelve times at the 100 km bandwidth, each time excluding one border pair from the sample, and compare the resulting coefficients to the pooled estimate in Table 16, with border pairs as rows and outcomes as columns.

For most outcomes and most pairs, the sign and significance pattern is stable: exclud-

ing any single one of eleven of the twelve pairs leaves every coefficient close to its pooled value, with the same significance stars. Ghana–Burkina Faso stands out as a high-leverage pair in the sample: dropping it alone moves the continuous education-years coefficient from 1.205^{***} to 0.120 (losing significance), and it noticeably shifts not-working upward as well. Reassuringly, education level, the coarser and arguably more reliable measure of schooling, remains significant even without this pair (0.125^{***}, against a pooled estimate of 0.334^{***}). So the qualitative finding that legal exposure raises educational attainment survives the exclusion of this pair; it is specifically the continuous years measure, which is more sensitive to a small number of high-leverage observations, that loses significance. Dropping any individual Namibia pair on its own moves estimates only modestly (consistent with the full four-pair exclusion in Table 12 requiring all four to be dropped together to materially change the result). The child-marriage coefficient, fertility outcomes, and age at first sex are stable across all twelve exclusions.

First-stage stability across outcome-specific samples Table 1's notes observe that the child-marriage coefficient varies marginally across outcome-specific estimation samples, since a small number of outcomes exclude some women (for example, age at first sex is defined only for women who have had sex). Table 17 reports that first-stage coefficient separately for every outcome-specific sample used elsewhere in the paper, rather than just the pooled value. For most outcomes the first-stage estimate is essentially unchanged from the pooled value of -0.172 in Table 1, ranging narrowly between -0.172 and -0.173 at the 100 km bandwidth (education years, any education, education level, children ever born, multiple unions, not working, and domestic work). Age at first birth, age at first sex, and education years conditional on schooling, whose samples are restricted to women who have already given birth, had sex, or received any schooling, respectively, show a somewhat smaller first stage (-0.158, -0.143, and -0.130 at 100 km). The white-collar samples are also somewhat smaller (-0.165 and -0.164). None of these deviations are large enough to suggest that the reduced-form contrasts in Table 1 are being driven by differential first-stage strength across outcome-specific samples rather than by the outcomes themselves.

Table 16: Leave-One-Border-Pair-Out: All Outcomes (100 km Bandwidth)

Pair	Child marr.	Educ. yrs	Educ. level	Age 1st birth	Children born	Age 1st sex	Mult. unions	Not working	Domestic wk	W-collar ext.	W-collar int.
All Pairs	−0.173***	1.205***	0.334***	0.931***	−0.330***	0.633***	−0.008	0.113***	−0.016***	0.029***	0.044***
BEN–BFA	−0.175***	1.229***	0.340***	0.949***	−0.338***	0.625***	−0.010	0.108***	−0.016***	0.031***	0.045***
BEN–NER	−0.172***	1.195***	0.334***	0.945***	−0.332***	0.643***	−0.008	0.109***	−0.016***	0.029***	0.043***
BDI–TZA	−0.185***	1.555***	0.381***	0.918***	−0.315***	0.656***	−0.012	0.137***	−0.021***	0.025***	0.043***
CIV–BFA	−0.180***	1.501***	0.401***	1.033***	−0.369***	0.682***	−0.007	0.106***	−0.017***	0.035***	0.051***
CIV–MLI	−0.175***	1.076***	0.327***	0.952***	−0.360***	0.617***	−0.001	0.114***	−0.016***	0.035***	0.048***
GHA–BFA	−0.127***	0.120	0.114**	0.538***	−0.162**	0.494***	−0.006	0.179***	−0.007	0.004	0.019
NAM–AGO	−0.156***	0.562***	0.238***	0.862***	−0.256***	0.539***	−0.005	0.067***	−0.021***	0.026***	0.037***
NAM–ZAF	−0.173***	1.208***	0.335***	0.928***	−0.336***	0.638***	−0.009	0.115***	−0.016***	0.029***	0.044***
NAM–ZMB	−0.177***	1.307***	0.356***	0.966***	−0.344***	0.648***	−0.011	0.101***	−0.017***	0.036***	0.048***
NAM–ZWE	−0.173***	1.211***	0.336***	0.937***	−0.330***	0.639***	−0.009	0.112***	−0.015***	0.029***	0.043***
RWA–TZA	−0.137***	1.032***	0.287***	0.796***	−0.223***	0.512***	−0.004	0.117***	0.000	0.023***	0.033***
UGA–TZA	−0.184***	1.293***	0.345***	0.975***	−0.372***	0.711***	−0.014*	0.119***	−0.017***	0.025***	0.038***

Notes: Each row reports the reduced-form coefficient on treated from a separate regression at the 100 km bandwidth, with the named border pair excluded from the sample ("All Pairs" is the unrestricted baseline, matching Table 1). Columns correspond to outcomes. All regressions include border-pair fixed effects, border-pair-specific slopes and kinks, geographic controls, ethnicity fixed effects, survey-year fixed effects, and individual controls, with standard errors clustered at the DHS-cluster level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 17: First-Stage Effect of Treatment on Child Marriage Across Bandwidths

	Bandwidth (km)		
Outcome sample	50	100	200
Education (years)	-0.141^{***}	-0.172^{***}	-0.159^{***}
	(0.022)	(0.018)	(0.015)
Any education	-0.141^{***}	-0.172^{***}	-0.159^{***}
	(0.022)	(0.018)	(0.015)
Education (years $\mid$ schooling > 0)	-0.108^{***}	-0.158^{***}	-0.154^{***}
	(0.030)	(0.024)	(0.019)
Education level	-0.142^{***}	-0.173^{***}	-0.159^{***}
	(0.022)	(0.018)	(0.015)
Age at first birth	-0.120^{***}	-0.143^{***}	-0.133^{***}
	(0.024)	(0.019)	(0.016)
Children ever born	-0.142^{***}	-0.173^{***}	-0.159^{***}
	(0.022)	(0.018)	(0.015)
Age at first sex	-0.089^{***}	-0.130^{***}	-0.129^{***}
	(0.023)	(0.020)	(0.016)
Multiple unions	-0.141^{***}	-0.173^{***}	-0.159^{***}
	(0.022)	(0.018)	(0.015)
Not working	-0.141^{***}	-0.173^{***}	-0.159^{***}
	(0.022)	(0.018)	(0.015)
Domestic work	-0.143^{***}	-0.164^{***}	-0.142^{***}
	(0.028)	(0.022)	(0.018)
White-collar (extensive)	-0.131^{***}	-0.165^{***}	-0.150^{***}
	(0.024)	(0.019)	(0.016)
White-collar (intensive)	-0.143^{***}	-0.164^{***}	-0.142^{***}
	(0.028)	(0.022)	(0.018)

Notes: Each row reports the coefficient on treated from a regression of the child-marriage indicator on treatment, estimated on the same outcome-specific sample used for the corresponding reduced-form regression in Table 1 (for example, the age at first sex row uses only women who have had sex). Standard errors clustered at the DHS-cluster level in parentheses. All regressions include border-pair fixed effects, border-pair-specific slopes and kinks, geographic controls, ethnicity fixed effects, survey-year fixed effects, and individual controls. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Sample selection in conditional outcomes Age at first birth and age at first sex are defined only for women who have, respectively, ever given birth or ever had sex by the time of the survey, so if legal exposure itself shifts the probability that these events have occurred, the timing effects reported in Table 1 could partly reflect a change in who is observed rather than a genuine delay. To bound this concern, we compute Lee (2009) bounds (Lee, 2009)

for both outcomes at the 100 km bandwidth using the `leebounds` Stata package (Tauchmann, 2014).

Table 18 reports the results. Selection into the conditional sample is somewhat higher on the control (lenient) side than the treated (strict) side for both outcomes—79.5 percent of control-side women have given birth versus 65.3 percent of treated-side women, and 77.6 percent versus 64.0 percent have had sex—consistent with earlier marriage on the control side giving more women time to reach these milestones by the survey date. Trimming the higher-selection arm to equalize selection rates yields bounds of $[0.934, 2.776]$ for age at first birth and $[0.995, 2.489]$ for age at first sex. Both intervals lie entirely above zero, so even under the most adverse admissible pattern of selection into the conditional sample, women on the strict side of the border still reach first birth and first sex significantly later than women on the lenient side. As the notes to Table 18 make clear, these bounds are computed on the raw treated–control contrast within the bandwidth, without the RD polynomial terms or fixed effects used in Table 1, so the bound width is not directly comparable to the reduced-form standard errors; the bounds establish the sign of the effect under worst-case selection rather than replacing the main point estimate.

Table 18: Lee (2009) Bounds for Conditional Outcomes (100 km Bandwidth)

Outcome	Selection (control)	Selection (treated)	Lower bound	Upper bound
Age at first birth	0.795	0.653	0.934	2.776
Age at first sex	0.787	0.632	0.995	2.489

Notes: Lee (2009) bounds for the timing outcomes, which are defined only on a treatment-affected subpopulation (ever gave birth; ever had sex). Selection columns report the share of women in each arm for whom the outcome is defined; the group with higher selection is trimmed from above/below to equalize selection rates, bounding the average treatment effect for always-selected women. Bounds are computed on the raw treated/control contrast within the 100 km bandwidth (Lee bounds do not accommodate the RD terms or fixed effects) and complement, rather than replace, the main RD estimates. $N = 70{,}178$ for both outcomes. Computed with the `leebounds` Stata package (Tauchmann, 2014).

Alternative ethnic fixed effects (GREG) The main specification uses ethnicity fixed effects from the digitised Murdock (1959) map of pre-colonial African peoples. As a robustness check on this choice, and following the analogous check in Anderson (2018), we replace the Murdock homelands with fixed effects from the Geo-Referencing of Ethnic Groups dataset (GREG; Weidmann, Rod and Cederman 2010), the digitized Soviet Atlas Narodov Mira. Each survey cluster is assigned the primary GREG group at its coordinate

(46 groups in the estimation sample, versus 102 Murdock homelands). Table 19 reports the results across bandwidths.

Every main result is robust to this alternative ethnicity classification. The child-marriage first stage, education, marriage- and fertility-timing, and labour-market coefficients are all of the same sign and significance as in Table 1, with magnitudes close to the main estimates despite the coarser partition of ethnic groups. Multiple unions remain insignificant, as in the main specification. This indicates the main results are not an artefact of the particular ethnic fixed-effects scheme chosen.

Table 19: Reduced-Form Effect of Treatment: GREG Ethnic Fixed Effects (Atlas Narodov Mira)

	Bandwidth (km)		
Outcome	50	100	200
Child marriage	−0.100***	−0.131***	−0.106***
	(0.024)	(0.018)	(0.014)
Education (years)	0.937***	1.097***	1.042***
	(0.243)	(0.202)	(0.170)
Education level	0.288***	0.326***	0.307***
	(0.044)	(0.037)	(0.031)
Age at first birth	0.680***	0.734***	0.506***
	(0.157)	(0.118)	(0.093)
Children ever born	−0.248***	−0.305***	−0.207***
	(0.071)	(0.056)	(0.045)
Age at first sex	0.390***	0.437***	0.242***
	(0.125)	(0.105)	(0.083)
Multiple unions	0.006	−0.013	−0.007
	(0.012)	(0.009)	(0.007)
Not working	0.079***	0.087***	0.125***
	(0.028)	(0.021)	(0.018)
Domestic work	0.005	−0.013***	−0.005
	(0.005)	(0.005)	(0.004)
Paid in cash	0.070*	0.048*	0.062***
	(0.036)	(0.026)	(0.022)
White-collar (extensive)	0.032***	0.034***	0.025***
	(0.009)	(0.007)	(0.006)
White-collar (intensive)	0.044***	0.049***	0.043***
	(0.012)	(0.010)	(0.008)

Notes: Identical to the main reduced-form specification (Table 1) except that the Murdock (1959) ethnic-homeland fixed effects are replaced with fixed effects from the Geo-Referencing of Ethnic Groups dataset (GREG; Weidmann, Rod and Cederman 2010), the digitized Soviet Atlas Narodov Mira, following the analogous robustness check in Anderson (2018). Each survey cluster is assigned the primary GREG group at its coordinate (46 groups in the estimation sample, versus 102 Murdock homelands). Standard errors clustered at the DHS-cluster level. $^{*}\,p<0.10$, $^{**}\,p<0.05$, $^{***}\,p<0.01$.

Two-way ethnicity × country clustering Table 20 reports two-way clustered standard errors at the ethnicity × country level, following Anderson (2018). They are uniformly larger than the DHS-cluster standard errors reported in Table 1, most starkly for education in years (0.612 versus 0.195 at the 100 km bandwidth), which points the same direction as the fragility already documented.

Table 20: Reduced-Form Effect of Treatment: Two-Way (Ethnicity × Country) Clustered Standard Errors

	Bandwidth (km)		
Outcome	50	100	200
Human capital			
Education (years)	0.962*	1.205**	1.298**
	(0.526)	(0.612)	(0.559)
Education level	0.282***	0.334***	0.334***
	(0.082)	(0.101)	(0.099)
Fertility			
Age at first birth	0.857***	0.931***	0.755***
	(0.134)	(0.121)	(0.100)
Children ever born	−0.266***	−0.330***	−0.280***
	(0.077)	(0.080)	(0.090)
Age at first sex	0.527***	0.633***	0.480***
	(0.158)	(0.123)	(0.146)
Household and labour market			
Multiple unions	0.010	−0.008	−0.011
	(0.007)	(0.008)	(0.010)
Not working	0.103**	0.113**	0.149***
	(0.046)	(0.045)	(0.041)
Domestic work	0.005	−0.016***	−0.008
	(0.007)	(0.006)	(0.007)
Paid in cash	0.045	0.057	0.067
	(0.046)	(0.041)	(0.041)
White-collar employment	0.026***	0.029***	0.026**
	(0.008)	(0.010)	(0.011)
White-collar among workers	0.033***	0.044***	0.044***
	(0.009)	(0.010)	(0.013)

Notes: Reduced-form estimates. The coefficient is the estimated effect of residence on the strict-law side of the border, matching Table 1. Standard errors in parentheses are two-way clustered at the ethnicity × country level via `ivreg2`, following Anderson (2018). With only fifteen countries and roughly one hundred ethnic homelands in the sample, many ethnicities are nested within a single country and the two-way cluster covariance matrix is not of full rank, so these estimates should be interpreted cautiously. All regressions include border-pair fixed effects, border-pair-specific slopes and kinks, pixel-level geographic controls, ethnicity fixed effects, and survey-year fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.5 RDD Graphs

The figures in this section present visual diagnostics for the regression discontinuity design. For the main outcomes, we plot binned means against signed distance to the relevant international border, with negative values denoting the control side and positive values denoting the treated side. The dots are constructed from twenty equal-observation bins on each side of the cutoff, so that each bin contains approximately the same number of observations within its side of the border. The solid lines show separate linear fits estimated on either side of the cutoff within the 100 km bandwidth. Because the figures pool multiple border pairs with different baseline outcome levels, they are intended as descriptive diagnostics rather than formal evidence. Inference is based on the reduced-form specification reported in the main text.

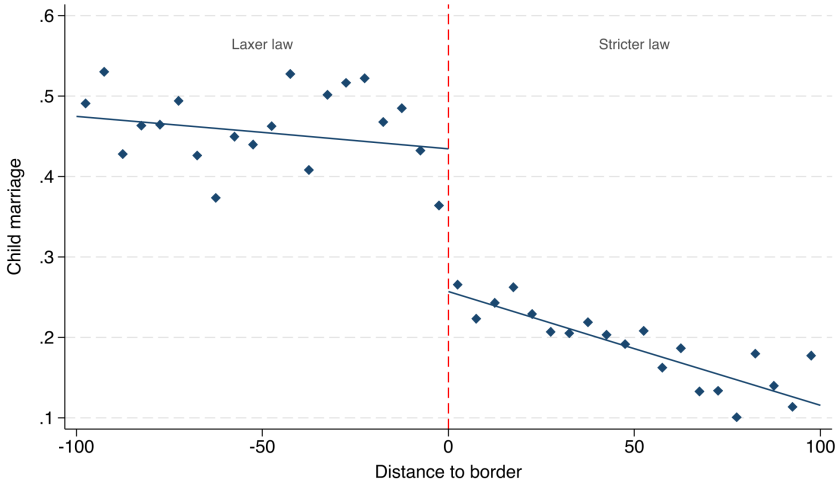


Figure 3: Effect on Child Marriage

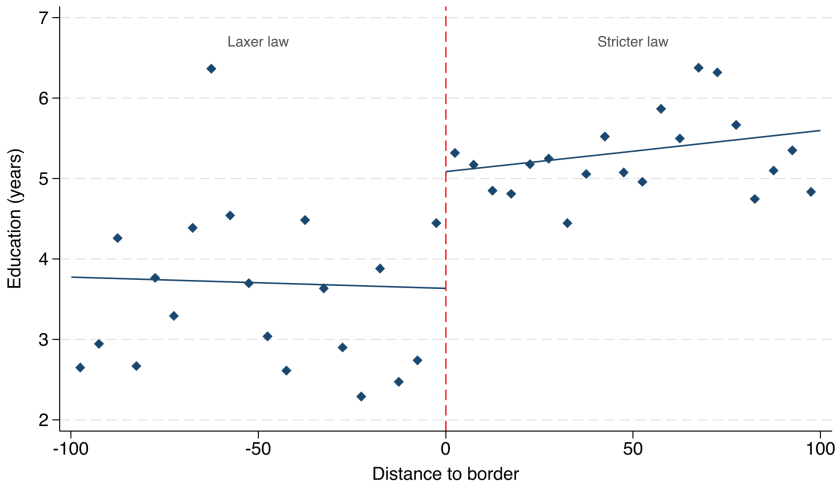


Figure 4: Education (Years)

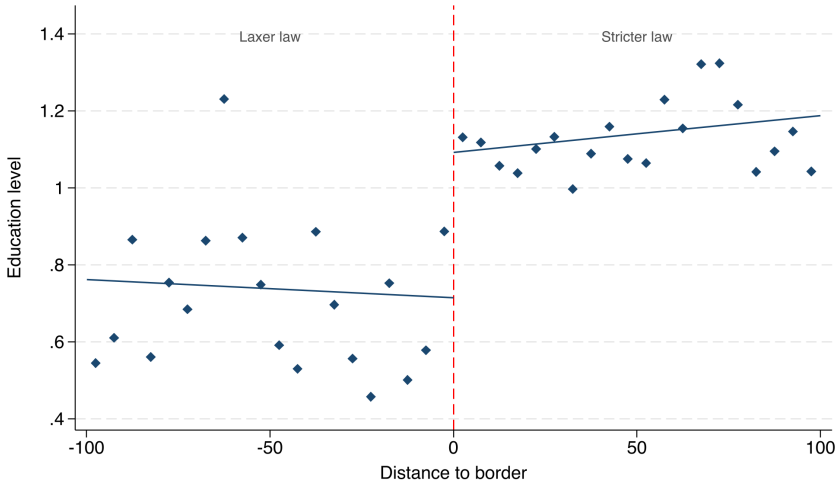


Figure 5: Education (Level)

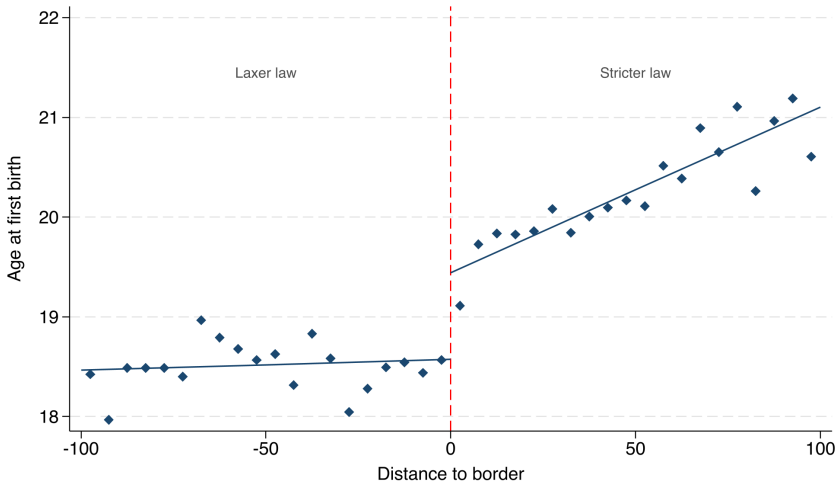


Figure 6: Age at First Birth

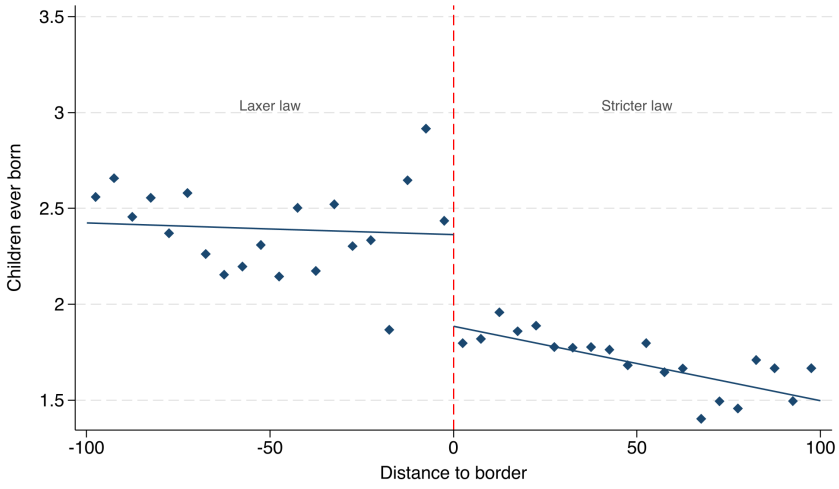


Figure 7: Children Ever Born

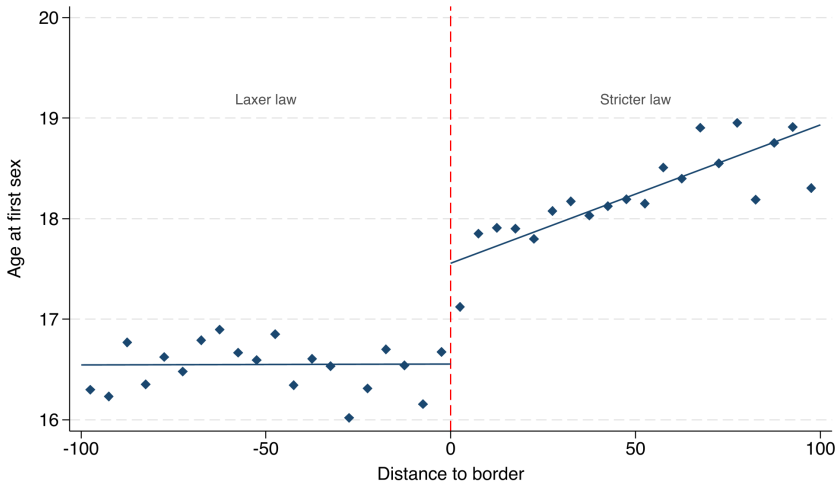


Figure 8: Age at First Sex

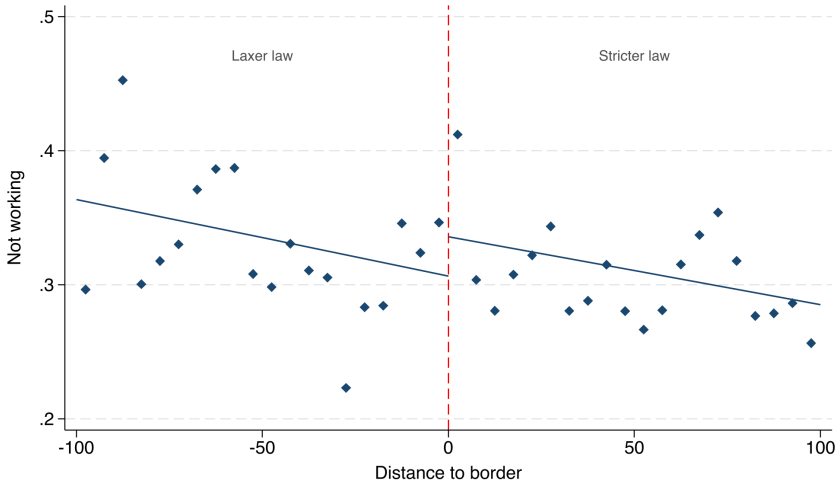


Figure 9: Not Working

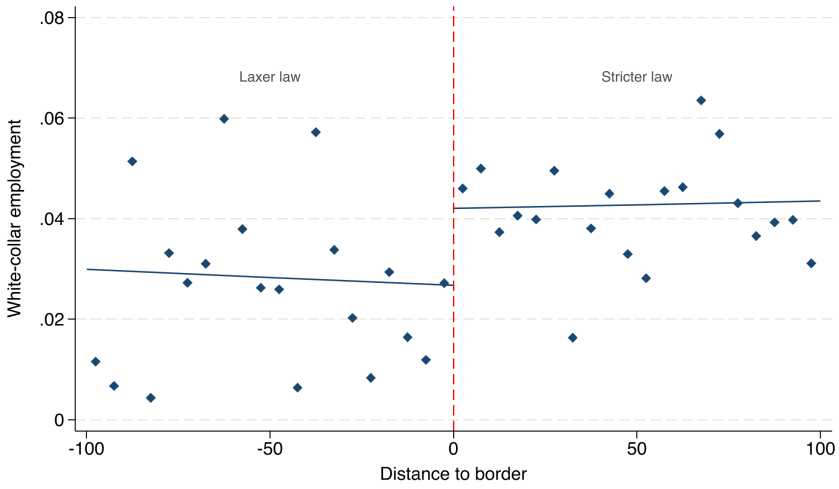


Figure 10: White Collar Extensive

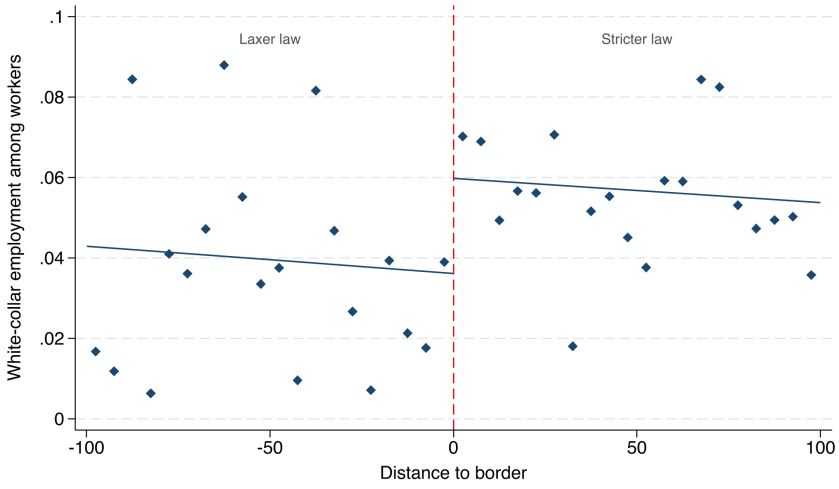


Figure 11: White Collar Intensive

A.6 Different Laws and Policy

Table 21: Cross-Border Legal Comparisons Across 12 Border Pairs

Law	Treatment Higher	Control Higher	Same
Age of Consent	2	2	8
Minimum Education	5	3	4
Women's Property Rights	5	2	5
Right to Work Outside Home	1	4	7
Right to Own a Bank Account	1	0	11

Notes: Each row compares the legal environment on the treatment side (higher minimum marriage age) against the control side across all 12 border pairs. "Same" indicates that both sides have identical legal provisions. All 12 pairs have complete data across all five indicators. Sources: World Bank Women, Business and the Law; WORLD Policy Analysis Center.

Using data from the World Bank's Women, Business and the Law indicators and the WORLD Policy Analysis Center, we compare the legal environment on the treatment and control sides of each border pair across five dimensions: age of consent, minimum years of compulsory education, women's property rights, the right to work outside the home, and the right to own a bank account. Table 21 summarises the results across all 12 border pairs.

The pattern is broadly balanced. For the right to own a bank account, the two sides are identical in 11 of 12 pairs. The right to work outside the home is the same in 7 of 12 pairs, with control-side countries having higher protections in 4 pairs. Women's property rights and minimum education show more variation, but neither is systematically higher on the treatment side. Age of consent is the same in 8 of 12 pairs. Taken together, these indicators suggest that treatment and control countries are not systematically different in their broader legal environment for women, reducing concern that the main estimates are driven by concurrent differences in women's rights outside the marriage-law channel.